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A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

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The Oxford Handbook of Educational Psychology

Edited by Angela O'Donnell

Subject: Psychology, Educational Psychology Online Publication Date: May 2018

DOI: 10.1093/oxfordhb/9780199841332.013.23

Abstract and Keywords

This article offers a retrospective and prospective analysis of the role of cognitive strategies in students' academic development over the past 25 years. The focus is on those processes that individuals employ to advance their own learning and understanding (learning strategies) and, to a lesser degree, those procedures applied to regulate and monitor that learning and understanding (metacognitive or self-regulatory strategies). Drawing on a groundbreaking review from 1988, the article examines how students' epistemic beliefs—their beliefs about knowing and knowledge—may affect strategic engagement. It also considers students' motivations or emotions that accompany learning and academic development and, therefore, strategic processing. Finally, it describes strategies associated with online learning, the barriers to being strategic in classrooms, and situations in which teachers and students can foster strategic thinking.

Keywords: cognitive strategies, academic development, learning strategies, metacognitive strategies, skills, epistemic beliefs, motivations, emotions, online learning, strategic thinking

In 1988, Alexander and Judy published a groundbreaking review that sought to bring together two long-established but distinct bodies of theory and research in cognitive psychology, specifically, the literatures on knowledge and strategic processing. As the authors stated at the outset of that critical analysis, there were two undeniable “truisms” about human learning and development that had long existed separately within the educational psychology literature, despite their seemingly symbiotic, if untapped, natures. The first truism was that learners' knowledge matters to subsequent learning and performance (e.g., Chi, 1985; Glaser, 1984), and the second was that those learners

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who are strategically engaged in the task at hand learn and perform better than those who are not so engaged (e.g., Flavell, 1981; Garner, 1987).

It was Alexander and Judy's contention that these two aforementioned "truths" about student learning and academic development could not be fully understood in isolation. Rather, any attempt to more fully explicate one of these truths demanded understanding of the other. Therefore, these authors set out to bring the research on knowledge and strategic processing into juxtaposition by deriving evidenced-based hypotheses about their interplay. They also set out to articulate an agenda for future research pertaining to the interaction of knowledge and strategies in students' academic development.

In framing the current examination of cognitive-based learning strategies, we have chosen to return to Alexander and Judy's (1988) often-cited review and interrogate its evidence-based hypotheses and recommendations for future research with two particular purposes in mind. First, retrospectively, we ask whether the issues and concerns outlined in that systematic review endure after more than 25 years of ensuing research, or whether, to the contrary, there has been noticeable progress in the area of cognitive-based learning strategies over the past quarter century. Second, prospectively, we look ahead to what the next 25 years may hold for strategy research and for those invested in student learning and development. In effect, are the recommendations for future research offered by Alexander and Judy in 1988 still pertinent or are there new avenues for inquiry that need to be explored?

What Are Cognitive Strategies Anyway?

Certainly, for a chapter devoted to cognitive strategies and academic development, we should legitimately begin with a clear explication of what we personally hold cognitive strategies to be. Thus, let us state from the outset that our focus in this discussion is on the mental or cognitive processes that individuals employ when they engage in problem-solving tasks. While there are various types and classes of cognitive strategies that can be addressed (and that we overview in a subsequent section of this chapter), our particular interest is on those processes that individuals employ to advance their own learning and understanding (Alexander, Schallert, & Hare, 1991; Weinstein & Mayer, 1986) and, to a lesser degree, those procedures applied to regulate and monitor that learning and understanding (Boekaerts, Pintrich, & Zeidner, 2000). The former are often referred to as learning strategies (Weinstein, Goetz, & Alexander, 1988), whereas the latter typically fall under the rubric of metacognitive or self-regulatory strategies (Zimmerman & Schunk, 2011).

As a consequence of this focus, we thereby preclude any consideration of instructional or pedagogical strategies from this examination. Instructional or pedagogical strategies are those procedures that educators employ to orchestrate a learning environment, influence students' behaviors or direct their attention, or promote learners' thinking or strategic

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behavior (Alexander, Graham, & Harris, 1998). Further, we refrain from any direct consideration of procedures that individuals may intentionally evoke to regulate their behaviors, motivations, or emotions (e.g., Beauregard, Lévesque, & Bourgouin, 2001) unless those procedures also involve learning and development in academic domains or in the context of schooling (e.g., Pekrun, Goetz, Titz, & Perry, 2002). In both these cases of exclusion, we do not question the importance or merits of such strategic endeavors or the research they represent. We simply cannot do justice to these areas as part of this exploration.

Regarding a more detailed characterization of strategies, we direct attention to the definitions of strategies and strategic processing that framed the Alexander and Judy (1988) review—definitions that have guided our empirical investigations over the ensuing decades (e.g., Alexander & Kulikowich, 1991; Alexander, Sperl, Buehl, Fives, & Chiu, 2004; Murphy & Alexander, 2002). Specifically, Alexander and Judy depicted strategies as a “special form of procedural knowledge that can exist in varying degrees of generality or separation” (1988, p. 376). The authors went on to state that strategies are “goal directed procedures that are planfully or intentionally evoked” and “aid in the regulation, execution, or evaluation” of a problem or task (p. 376). Such a conceptualization mirrored definitions of cognitive strategies forwarded by scholars contributing to that expanding literature in the 1970s and 1980s (e.g., Campione & Brown, 1978; Weinstein & Mayer, 1986).

Interestingly, while Alexander and Judy (1988) expressed concern about the clarity with which the terms *domains* and *domain knowledge* were conceptualized in the existing empirical literature of the time, they seemingly operated under the assumption that the definition of strategies they proffered was rather noncontroversial. This may have been an unwarranted assumption on the authors’ part, given the conceptual quandary that has since come to light within contemporary research and instructional practice (Afflerbach, Pearson, & Paris, 2008), namely, the differentiation between skills and strategies.

Conceptual Quandary: The Boundaries Between Skills and Strategies

The distinction between strategies and skills has seemingly blurred over time—a concern strongly expressed by Alexander et al. (1998) and others (Afflerbach et al., 2008; Alexander, 2006). Specifically, as Alexander et al. (1998) lamented 10 years after the Alexander and Judy (1988) review:

Despite the value that practitioners and the public, at large, seemingly place on basic skills or “skillful” behavior, efforts to juxtapose conceptions of strategic and skillful performance have been relatively rare in the literature. (p. 130)

Contributory Factors

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Definitional confusion.

It is one thing to acknowledge that a quandary exists between conceptions of strategies and skills in human learning and academic development that mark educational discourse. It is quite another to understand *why* they exist or *how* to address such problems. Of course, some of the more obvious but simplistic explanations for this continuing conceptual problem can be laid at the feet of educational researchers and practitioners who are careless in the language they use or negligent in providing explicit definitions of terms they employ (Alexander et al., 1991; Murphy & Alexander, 2000). However, there are more complex and problematic reasons that must be forwarded.

For one, this conceptual problem may, in part, be a side effect of the regularity with which students' strategic abilities and strategic processing are touted within educational policies (Common Core State Standards Initiative [CCSS], 2010) or instructional practice (Afflerbach & Cho, 2008, 2010)—a promotion that too often occurs in the absence of explicit definitions of key terms (Alexander et al., 1991; Murphy & Alexander, 2000). As a result, the more the term *strategies* has come to populate the educational discourse, the less its meaning vis-à-vis skills has been effectively communicated or embodied (Alexander et al., 1998). This lack of clarity is more than a semantic dilemma, but comes with serious instructional and assessment consequences, as Afflerbach et al. (2008) have emphatically contended.

Intentionality and effortfulness.

Another contributor to the fact that the terms *strategies* and *skills* are often used interchangeably in the literature (Alexander et al., 1998) may be that these two critical forms of process do share certain attributes. For instance, both represent forms of procedural knowledge essential for human learning and academic performance that must be acquired and honed over time. Nonetheless, there are significant differences between strategies and skills that should not be ignored. As Alexander et al. (1998) noted:

These differences pertain to the automaticity of performance and to learner awareness or intentionality. ... In effect, skills are procedures that have been routinized. That is, students hone these techniques to a level of automaticity, enabling them to perform a given task fluidly and effectively. Skills are the "habits" of performance, or what a learner typically does (James, 1890). ... When students' routine techniques or typical approaches are not enough to resolve a given situation, conscious, intentional, and effortful processing must take over. ... When this conscious, intentional, and effortful processing takes over, the learner is performing strategically. (p. 135)

Surprisingly, when we fast forward 20 years since Alexander and Judy's (1988) article, we find that this confounding of skills and strategies remains problematic, as evidenced by the arguments articulated by Afflerbach et al. (2008) in an article intended primarily for reading practitioners. As these authors stated in the abstract to this article:

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There is a lack of consistency in the use of the terms skill and strategy, reflecting an underlying confusion about how these terms are conceptualized. Such inconsistency can confuse students and teachers and render instruction less effective. (p. 364)

In concert with Alexander et al. (1998) and Alexander and Judy a decade earlier, Afflerbach et al. (2008) sought to resolve the confusion they identified in the reading literature by establishing that

it is important that the terms *skill* and *strategy* be used to distinguish automatic processes from deliberately controlled processes. (p. 371)

But, this seeming failure to recognize the conscious, intentional, and effortful character of strategies is not solely a problem of educational practice. Rather, it is reflected in the empirical research that attempts to investigate learners' strategic processing but that does not seem to disentangle such "deliberate and controlled processes" (Afflerbach et al., 2008, p. 371) from the habituated or automatic processing in which students engage (Campione & Brown, 1978). This remains problematic even in studies by researchers (ourselves included) who conceptually acknowledge this critical distinction between skills and strategies but find it challenging, if not impossible, to distinguish automatic or habitual processes applied to some cognitive task from those that have a more conscious and effortful character (Dinsmore & Alexander, 2012).

This discrepancy between the conceptualization and operationalization of strategies in much educational research may arise, in part, from the methods commonly employed to unearth cognitive processes during task performance, such as think-alouds (Pressley & Afflerbach, 1995). As has long been argued, such methods, while invaluable for transforming internal or tacit thoughts into external and explicit data, inevitably alter the very processes being investigated (Ericsson & Simon, 1998; Winne & Perry, 2000). The consequence of such transformations for the study of cognitive strategies (versus skills) is especially noteworthy. For instance, consider when students are called on to verbalize whatever they are thinking or doing when cognitively engaged. Researchers analyzing such think-aloud data cannot hope to definitively determine whether the voiced processes are evidence of respondents' "default" or skillful manner of operation or indicative of their more intentional actions. Of course, the degree to which think-alouds or other data-gathering procedures (e.g., retrospective or structured interviews) become intrusive can be carefully addressed in their implementation (Pressley & Afflerbach, 1995). Because of the importance that data-gathering procedures play in the study of cognitive strategies, we revisit this issue of research methodologies when we consider future directions by exploring alternative or complementary research techniques.

Pedagogical concerns.

Yet another contributory factor in this conceptual confusion may have a more pragmatic and pedagogical root. Specifically, as theoretically established, a critical distinction between strategic and skillful processing rests in the degree of automaticity or

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deliberateness each conveys. Thus, when educators teach procedures for predicting, summarizing, or clarifying, it may not be evident at the time whether they are teaching procedures that will remain intentional and effortful for those students (i.e., strategies) or eventually reach automaticity for them (i.e., skills). In other words, the ultimate distinction in this case will reside in the way in which these essential procedures are internalized and activated by the learners themselves. Because educators must operate in the dark to some extent, it may be understandable that their use of terms like *strategies* or *skills* to denote these cognitive procedures remains obscured. We revisit issues pertaining to the teaching of strategies later in this chapter.

Contributions of Epistemic Beliefs and Motivation/Emotion Research

Because human intentionality and effortfulness are cast as so pivotal to the aforementioned distinction between strategies and skills, there are emerging bodies of research that would seem to be particularly relevant to this discussion. The first pertains to the beliefs about knowing and knowledge that students hold and the effect that those beliefs may exert on strategic engagement (Murphy, Alexander, & Muis, 2012; Murphy & Mason, 2006). For instance, because knowledge is a centerpiece of this partnership, individuals' beliefs about knowledge and knowing—epistemic beliefs—are naturally entering the landscape in strategy theory and research (Muis, 2007; Sinatra & Pintrich, 2002). Thus, it seems reasonable to assume that decisions to engage in conscious, deliberate, and effortful procedures to promote learning and performance would be influenced by students' beliefs regarding whether knowledge is simple or complex or whether it is bestowed by authorities or derived via critical analysis and justification (Alexander et al., 2011; Schommer, 1993).

The second encompasses students' motivations or emotions that accompany learning and academic development and, therefore, students' strategic processing (Linnenbrink-Garcia & Pekrun, 2011; Pekrun et al., 2002). Indeed, the research on motivations and emotions represents two of the most fertile lines of inquiry to take form since the publication of the Alexander and Judy article. Acknowledging the significance of these lines of inquiry to the nature of strategic (vs. skillful) processing, the Alexander and Judy wrote:

Because strategies, as we have operationally defined them, are characterized by planful and intentional use, [strategies'] susceptibility to motivational effects may be rather considerable. ... One recent volume, devoted specifically to the topic of metacognition, motivation, and understanding offers an extended discussion of the relationship between strategy use and motivation. (p. 396)

Because we cannot help to do justice to these two domains of inquiry, we only briefly acknowledge their roles as contributors to the distinctions between strategies and skills. We leave it to others, including contributors to this volume, to extend this discussion.

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With this conceptual overview in place, we now turn to other specifics about cognitive strategies and student learning and performance. Specifically, within the retrospective frame, we explore the various types and levels of strategies suggested by Alexander and Judy's (1988) review, especially with regard to the topics of domain-general versus domain-specific and higher order versus lower order strategies. Through the prospective lens, we then query the literature to identify new manifestations of strategic processing that may merit increased attention in the decades to come. Similarly, reaching back into the literature, we revisit concerns over the failure to foster and facilitate students' strategic thinking and performance within the context of schools. Then, we peer ahead to consider general and specific steps that can be taken within schools and classrooms to bolster the likelihood of effective strategic processing by students who populate those educational environments.

Types of Strategies

As noted, Alexander and Judy (1988) identified the lack of clarity within the educational and psychological literature regarding what constitutes a strategy as a substantial roadblock for strategy research. Of particular concern was the impact of this lack of clarity on investigations that examined the relation between strategic processing and domain knowledge. Specifically, Alexander and Judy posited the recommendation that "future investigations should provide clear operational definitions and detailed descriptions of the subjects, domains, and strategies involved" (p. 397). They also argued that the conceptual confusion arising from unclear operational definitions "seems particularly acute when we try to discern the relationship of more domain-specific strategies to those of a more general or executive nature" (p. 382). Given the focus of Alexander and Judy's 1988 review on the relation between domain knowledge and strategic processing, the question of the domain specificity of strategies was essential. In the 25 years since, the relation between domain knowledge and strategy use has been well established in the empirical literature (e.g., Alexander et al., 1998; E. Fox, 2009; Moos & Azevedo, 2008). Yet, has the distinction between domain-specific and domain-general strategies been clarified in the intervening years?

Moreover, the interplay between strategies and knowledge extends beyond questions of domain generality or specificity to concerns about the depth and level of strategic processing. Thus, we turn our attention to a discussion of three specific classes of strategies that would signify not only a relation between cognitive strategies and knowledge but also academic development more broadly: (a) domain general and domain specific, (b) deep processing and surface processing, and (c) cognitive and metacognitive. Our primary purpose is to interrogate the literature to determine whether we have achieved the greater sense of clarity called for by Alexander and Judy (1988).

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However, before we embark on this endeavor, we forward one critical caveat. As with distinctions between and among constructs that populate the educational psychological literature, there are rarely clean or truly dichotomous relations between classes of strategies—regardless of how we choose to categorize them. It may be advantageous to set deep against surface, for instance, or domain general against domain specific to sharpen contrasts between such classes of strategies. But, reality is that there is little justification for any true dichotomization in these instances. The distinctions that may exist are potentially more continuous as opposed to discrete in nature. Further, marked differences in the nature of learners, tasks, or contexts can reposition any specific problem-solving procedure along its defining continuum. Thus, what qualifies as a surface-level strategy for Learner A performing Task B in Context C would more accurately qualify as a deep-processing strategy for Learner X engaged in Task Y within Context Z. With this in mind, we address the challenge of conceptual clarity, confronting whether and how those types of strategies have been defined and examined within the literature, and whether the “lack of understanding” (p. 382) described by Alexander and Judy (1988) remains.

Domain-General and Domain-Specific Strategies

For Alexander and Judy (1988), the need to understand whether strategies are specific for tasks within a single domain or whether they generalize across domains was critical. They noted:

To some, strategies are general processes that operate across domains ... whereas to others they are compilations or extensions of domain-specific knowledge. (p. 382)

Because they found that researchers tended to hold either a domain-general or a domain-specific view of strategies, Alexander and Judy subsequently discussed these classes of strategies as related, highlighting the importance of clarification in order to examine the relation between domain-general and domain-specific strategies. To this end, they pointed to a hierarchical framework for understanding distinctions between domain-general and domain-specific strategies forwarded by Pressley, Goodchild, Fleet, Zajchowski, and Evans (1989).

In this framework, Pressley, Goodchild, and colleagues (1989) identified several main categories of strategies that differ in their ability to generalize across domains (i.e., specific field of study; Muis, Bendixen, & Haerle, 2006). These range from strategies that are used to accomplish domain-specific tasks (e.g., mnemonics to remember the lines of a musical staff) to those that can be applied across many domains (e.g., activating prior knowledge). Domain-general strategies are strategies that can be applied to tasks and problems in a wide variety of content areas. These include cognitive strategies such as rereading, questioning, and idea generation (Murphy & Alexander, 2002; Pressley & Afflerbach, 1995; Youssef, Ayers, & Sweller, 2012), in addition to metacognitive strategies

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of planning, regulating behaviors, and monitoring comprehension (Nett, Goetz, Hall, & Frenzel, 2012; Pintrich, 2000; Zimmerman, 1995, 2002). Domain-general strategies can be traced back to research in reading, problem-solving, and memorizing, and they tend to rely on heuristics (Edwards Olshavsky, 1976; Flavell, Friedrichs, & Hoyt, 1970; Garner, 1982; Weir, 1964). In contrast, domain-specific strategies are those instantiated in a single domain, for the purpose of engaging in the tasks, activities, and goals specific to the domain at hand (Pressley, Goodchild, et al., 1989; Sweller, 1990). Domain-specific strategies are frequently algorithmic (Case & Marshall, 2004) and include common examples such as the five-paragraph sandwich model for essay writing, the FOIL (first, outer, inner, last) method in algebra, and the use of roman numerals to identify chord progressions in music.

In the past 25 years, perhaps the most widely researched domain-general strategies are those related to reading (e.g., Afflerbach, 1990; Cerdán & Vidal-Abarca, 2008; Murphy & Alexander, 2002). While certain reading-related strategies are domain specific because they are tied directly to the text (e.g., figuring out the meaning of a word in context, adjusting the rate of reading), many strategies related to reading comprehension are of a more general character. Given the nature of reading as a means for learning in a wide variety of domains (Alfassi, 2008; Shanahan & Shanahan, 2008), and given the historical roots of prominent reading researchers in the study of strategic processing (e.g., Garner, 1982; Pressley, Johnson, Symons, McGoldrick, & Kurita, 1989; van Dijk & Kintsch, 1983), it is no surprise that these strategies have been prominently featured in the domain-general strategy literature. As the context in which students engage in reading has shifted monumentally in the past 25 years, so has the focus on reading-related strategies. In this time, research on strategies related to searching for, selecting, and evaluating texts, as well as integrating across multiple texts, has been introduced and has begun to receive greater attention (Anmarkrud, Bråten, & Strømsø, 2014; Kurby, Britt, & Magliano, 2005; Wineburg, 1991). We return to the discussion of online learning strategies later in this chapter.

Whether domain specific or domain general, strategies do not operate without a context, and findings related to the success of domain-general relative to domain-specific strategies are contingent on understanding that context (Afflerbach & Cho, 2008). Yet, is the domain in which strategies are applied made evident in strategy research? Alexander and Judy noted that much of the general strategy research offered “at best, a weak articulation of the content area in which the strategy or strategies were applied” (1988, p. 380). In addition to a lack of identification of domains, they suggested that the domain in which strategies are examined should be purposeful. They argued, “It would also facilitate our understanding of the content-strategy relationship to have some information as to *why* particular domains were selected for investigation” (1988, p. 381). Such domain considerations are important, given the critical nature of domain differences outlined by Alexander and Judy (1988) and others (Afflerbach & Cho, 2008; Alexander et al., 1998).

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In 25 years, how well has strategy research met this call for clarity? In a 2012 review of deep and surface processing, Dinsmore and Alexander found that approximately 40% of studies did not situate strategic processing within a domain. This is not surprising given the directions of many strategy questionnaires, such as the Motivated Strategies for Learning Questionnaire (MSLQ; Pintrich, Smith, Garcia, & McKeachie, 1993) and the inventory of learning processes (ILP; Schmeck, Ribich, & Ramanaiah, 1977) to report general dispositions toward strategic processing. Samuelstuen and Bråten (2007) argued that such self-report global strategy measures have limited validity evidence. In this argument, they identified challenges both to students' ability to provide accurate judgments regarding their strategy use and limited criterion validity between global strategy measures and performance on academic tasks. As such, they suggested that adapting measures to focus responses on task-specific instances provides a more accurate measure of strategy use and serves as a better predictor for task performance. Breaking domain-specific strategies into task-specific strategies provides a more specific context for the consideration of strategic processing (Hagan, Braasch, & Bråten, 2014). Samuelstuen and Bråten's (2007) position harkens back to Pressley, Johnson, et al. (1989) conceptualization of domain-specific strategies as strategies limited to applications in a limited number of tasks within a domain.

Since the publication of Alexander and Judy's 1988 review, the importance of understanding the role of domain-specific versus domain-general strategies within a given context has been noted by a few studies that specifically addressed such issues (Pokay & Blumenfeld, 1990). For instance, in a study of students in a high school geometry course, Pokay and Blumenfeld (1990) found that reported use of domain-specific and domain-general strategies was related; however, strategies specific to geometry predicted achievement in the course, but general strategies did not. This is not to suggest that domain-general strategies do not support learning. Consistent with the second truism presented in Alexander and Judy's 1988 examination of strategic processing and domain knowledge, it has been well documented that students who use domain-general strategies have increased learning (e.g., Chinnappin & Lawson, 1996; Pape & Wang, 2003; Youssef et al., 2012).

Moreover, the goals of education to support higher order or critical thinking suggest that domain-general strategies should be the standard for developing students as strategic individuals. For students, the ability to apply strategies across a number of situations and domains not only is more efficient than learning domain-specific or task-specific strategies, but also is a hallmark of higher order thinking (Alexander et al., 2011; Resnick, 1987). Given concerns with measurement of domain-general strategies (Samuelstuen & Bråten, 2007) and the need to situate studies of strategic processing within carefully selected domains (Alexander & Judy, 1988), research examining the application of general cognitive strategies across tasks and domains will be necessary for understanding the degree to which these strategies relate to academic performance and to more accurately assess their relation to domain-specific strategies.

Deep-Processing and Surface-Processing Strategies

In addition to the division of strategies as domain specific or domain general, for the past 40 years, the educational psychology literature has also distinguished strategies in terms of the depth of processing that strategies entail (Marton & Säljö, 1976a, 1976b). The presence of this distinction has increased in empirical research since that time (Dinsmore & Alexander, 2012). Surface-processing strategies are defined as strategies used to make sense of the problem at hand (Dinsmore & Alexander, 2012; Nolen, 1988). In contrast, deep-processing strategies are used to integrate, transform, or evaluate a problem (Alexander et al., 2005; Nolen & Haladyna, 1990). Surface-processing strategies include rereading, underlining, and rehearsing points to remember (Phan, 2009; Pressley & Afflerbach, 1995; Rodriguez, 2009), while deep-processing strategies include elaborating on main ideas, identifying an underlying argument, and providing justification for a claim (Alexander, 2003; Chin & Brown, 2000; Nolen, 1988; Pressley & Afflerbach, 1995).

One of the first models to differentiate levels of strategic processing was the approaches to learning model developed by Marton and Säljö (1976a, 1976b) and modified by Biggs (1987). In their model, Marton and Säljö (1976a) regarded deep processing and surface processing as approaches to learning that vary across individuals (1976a) and across tasks (1976b). Through a series of studies, they suggested that some individuals are more likely to solve problems using deep-processing strategies, while others might be more inclined to use surface-processing strategies (Marton & Säljö, 1976a). Additionally, although noting that the type of questions that are asked elicit differences in strategies, they maintained their focus on strategic processing as a style of learning that differs across individuals, indicating that certain types of questions result in a deep-processing approach while others result in a surface-processing approach. Marton and Säljö's (1976a, 1976b) conceptualization was based on reading research with undergraduates, and as such, in their model, surface processing was concerned with strategies related to the replication of text, while deep processing was concerned with strategies for understanding the main ideas and principles of text. Along a similar framework to Marton and Säljö (1976a, 1976b), Biggs (1987) identified three levels of processing approaches, focusing on undergraduates' study habits. His levels ranged from the most surface-level processing (i.e., reproducing), to a moderate level (i.e., internalizing), and one consistent with deep-level processing (i.e., organizing).

In contrast to the approaches of learning model, information-processing models have used levels of processing to explain the encoding and storage of memories (Craik & Lockhart, 1972; Craik & Tulving, 1975; Kapur et al., 1994). The information-processing framework describes memories as moving from sensory to short-term to long-term storage (Baddeley, 1992). During this process, levels of processing are used to explain the degree of analysis individuals exercise to make sense of sensory input and information stored in short-term and long-term memory (Craik & Lockhart, 1972). In this model, surface-processing strategies include rehearsing information, while deep-processing strategies include elaboration and connection to prior knowledge (Craik & Tulving, 1975).

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The level of processing is regarded as varying according to the meaningfulness of the information presented, level of attention paid, and degree of overlap with current cognitive schemas (Craik & Lockhart, 1972). Accordingly, deep-processing strategies would most likely occur if the information is presented meaningfully, carefully attended, and aligns closely with an individual's existing schemas. Similar to the approaches to learning model, deep-processing strategies in the information-processing approach are associated with more efficient and effective retention of memory than surface-processing strategies.

Building on earlier models that differentiated deep and surface processing, Alexander created the Model of Domain Learning (MDL; Alexander, 1997, 2003), which situated deep and surface processing within the changes that occur across academic development and described developmental trajectories of deep- and surface-processing strategies. Unlike the approaches to learning model, the MDL provides for the concurrent use of surface- and deep-processing strategies within an individual and focuses on the interrelated changes in knowledge, motivation, and strategic processing that occur across development. In her MDL, Alexander (1997, 2003) suggested that as individuals move from the earliest stages of development within a domain (i.e., acclimation) to the more advanced stages of competence and proficiency, they tend to rely less on surface-processing strategies and more on deep-processing strategies. In 1988, Alexander and Judy noted that "as knowledge in a domain increases, strategic processing is altered" (p. 391). They discussed differences between novices and experts, alluding to differences in surface-processing and deep-processing strategy use between these groups. While in 1988 there was a tendency to believe that deep strategies are better than surface, Alexander and Judy (1988) indicated that the level of strategic processing required to complete a task may necessarily depend on the characteristics of the learner. The MDL built on research on expertise, allowing for a continuum of development and taking into account the differences in domain knowledge that were central to Alexander and Judy (1988)'s review.

In the past decades, certain misconceptions about deep- and surface-level strategic processing have emerged in the literature, misconceptions that present challenges to the task of reaching clarity. First, deep- and surface-processing strategies have been bifurcated, often with the belief that deep-processing strategies are always better than surface-processing strategies (Beattie, Collins, & McInnes, 1997; Dinsmore & Alexander, 2012). Similar challenges to the view of higher levels of processing as superior to lower levels have been brought forward with regard to higher order and lower order thinking (Alexander et al., 2011). However, despite the common belief by both researchers and students that deep-processing strategies are better for learning than surface-processing strategies (Nolen, 1988), empirical research has provided inconsistent findings (Block, 2009; Cano, 2007).

Dinsmore and Alexander (2012) credited this inconsistency to a lack of consideration for the context of learning. Moreover, the interaction of individual factors with strategic processing suggests that deep-processing strategies should not be expected and may not

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be conducive for all individuals across all contexts. For instance, an individual's domain knowledge and goals are important considerations for understanding the types of strategies that can be applied (Afflerbach, 1990; Coiro & Dobler, 2007; Phan, 2009). Just as Alexander and Judy (1988) thoroughly examined the interaction between strategies and domain knowledge, it will be important for research moving forward to examine the interaction between levels of processing and the contexts and situations in which deep, surface, or a combination of strategies best support learning.

The second misconception is that deep-processing and surface-processing strategies are frequently perceived as orientations to learning (Entwhistle, 1991; Marton & Säljö, 1976a) rather than part of a repertoire of strategies that individuals can apply across varied contexts and domains. In this sense, individuals are considered to be deep processors or surface processors (e.g., Verheij, Stoutjesdijk, & Beishuizen, 1996). Perceiving strategy use as an approach to learning is consistent with Marton and Säljö's (1976a) initial conceptualization; however, as they and others later identified (Marton & Säljö, 1976b; Ramsden, 1997), the task in which individuals engage can alter whether deep-processing or surface-processing strategies are most advantageous. Empirical research has found that individuals tend to use both deep-processing and surface-processing strategies during tasks such as reading (Pressley & Afflerbach, 1995; Afflerbach & VanSledright, 2001; Hagan et al., 2014) and learning (Aharony, 2006; Dunlosky, Rawson, Marsh, Nathan, & Willingham, 2013). Additionally, models such as the MDL have suggested that while individuals may have a propensity to use particular levels of strategies depending on their stage of academic development within a domain, these strategies work in conjunction and vary according to the domain in which individuals engage.

Returning to Alexander and Judy's (1988) call for strategies to be clearly defined, it is important to consider whether research on surface-processing and deep-processing strategies has met this goal. A recent review of deep- and surface-processing strategies by Dinsmore and Alexander (2012) identified this as still problematic and implicated it as a problem that seems to be growing as interest in research on levels of strategic processing has become more frequent since the 1980s. In their systematic review, Dinsmore and Alexander (2012) noted changes in the occurrence of definitions of deep- and surface-processing strategies over time. While definitions were provided for all but one article prior to 2000, a substantial proportion of articles published after 2000 did not include explicit definitions, a finding they suggested may be due to the increased prevalence of research in this area leading researchers to believe that definitions may not be necessary. However, as Alexander and Judy (1988) noted and as was evident in our previous discussion of strategies versus skills, the increase in research does not indicate that definitions should be discarded, rather that clarity should become more important as separate lines of inquiry are brought together.

Cognitive and Metacognitive Strategies

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Despite the specific focus of this chapter on cognitive learning strategies, we find it essential to recognize an additional class of strategies that are frequently considered in tandem with cognitive strategies: metacognitive strategies. In 1987, Ruth Garner devoted space in her book on *Metacognition and Reading Comprehension* to explaining the somewhat confusing association between cognitive and metacognitive strategies. The deceptively simple differentiation she offered was the enactment of mental procedures to rectify or enhance performance (cognitive strategies), the oversight or monitoring of those procedures (metacognitive strategies), or the overarching knowledge and experiences individuals have about their strategic capabilities (metacognitive knowledge and metacognitive experiences). In his seminal writings on metacognition, Flavell (1979) also acknowledged the challenge of separating cognitive and metacognitive strategies, acknowledging that the same strategy could be used to meet the cognitive goals of rectifying or enhancing performance as well as the metacognitive goals of overseeing or monitoring that performance.

Metacognitive strategies are frequently regarded as guiding individuals' decisions about their use and implementation of cognitive strategies (Flavell, 1979; Pressley, Johnson, et al., 1989; Purpura, 1997; Vrugt & Oort, 2008). As such, metacognitive strategies are intertwined with cognitive strategies, and the reciprocal process between metacognition and cognition presents a challenge for defining and measuring the strategies separately (Veenman, Van-Hout Wolters, & Afflerbach, 2006). Nonetheless, in recent studies, cognitive and metacognitive strategy use have been found to be distinct, yet positively related, to performance for tasks such as visuospatial construction (Dermitzaki, Leondari, & Goudas, 2009); reading comprehension (Yang, 2012); and second-language learning (Hsiao & Oxford, 2002). Moreover, as the interplay between domain knowledge and cognitive strategic processing has been established (Alexander et al., 1998; Alexander & Judy, 1988; Harris, Alexander, & Graham, 2008), so has the relation between metacognitive strategies and knowledge, especially in the expanding literature on experts and expertise development (Schraw, 1998). In brief, those who demonstrate greater expertise in a given domain or with particular problem-solving tasks likewise display greater metacognitive knowledge and strategic behavior over those who are less expert (Alexander et al., 2004; Ericsson, Krampe, & Tesch-Römer, 1993).

Over the past quarter century, there has been much less attention paid to the cognitive versus metacognitive distinction and much more dialogue directed to the relation between metacognitive and self-regulatory strategies, which both involve monitoring or regulation (Azevedo, 2009; E. Fox & Riconscente, 2008; Schunk, 2008). Given the growing presence of theory and research on self-regulation over the past 25 years (Schunk, 2008; Zimmerman, 1995, 2002), this debate over what constitutes metacognitive versus self-regulatory strategies shows no signs of abating (Dinsmore, Alexander, & Loughlin, 2008). However, it is beyond the scope and intention of this chapter to seek to tease apart the complex relation between metacognitive and self-regulatory strategies.

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Thankfully, there are others who have undertaken this difficult challenge (Dinsmore et al., 2008; Schunk, 2008; Zimmerman, 1995).

Measuring Strategies

Since the publication of Alexander and Judy's (1988) review, strategies and strategic processing have had a history of being measured via self-report questionnaires (e.g., Alexander & Murphy, 1999; Pintrich & De Groot, 1990); assigning conditions (e.g., Ally, Gold, & Budson, 2009; Stein & Kirby, 1992); think-alouds (e.g., Anmarkraud et al., 2014; Pressley & Afflerbach, 1995); and eye tracking (e.g., L. Mason & Ariasi, 2011; Moeller, Klein, & Nuerk, 2011), with some work in neurophysiological methods (e.g., Cho, Ryali, Geary, & Menon, 2011; Grabner et al., 2009). With the exception of neurophysiological methods, the other types of measures were present a quarter century ago. However, the years since have brought about certain transformations in the popularity of particular measures as well as advances in technology, which have aided the ability of researchers to collect real-time measures of strategic processing.

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Questionnaire measures and assigned conditions.

Common strategy self-report questionnaires include the MSLQ (Pintrich et al., 1993); the Study Process Questionnaire (SPQ; Biggs, 1987); and the ILP (Schmeck et al., 1977). These questionnaire measures direct individuals to report their typical manner of processing, aligning more closely to theoretical models by Marton and Säljö (1976a) and Biggs (1987) and focusing on domain-general strategies. While these measures have a long history of use in educational research with established psychometric properties (Pintrich & De Groot, 1990; Pintrich et al., 1993; Schmeck et al., 1977) and have been found to relate to academic performance (Eshel & Kohavi, 2003; Vrugt & Oort, 2008), they present challenges for researchers examining strategy use as a flexible, goal-directed process that varies across tasks and contexts. Some researcher-developed questionnaire measures have addressed this issue of task and context specificity (Bråten & Strømsø, 2011; Schellings, 2011), and others have applied more general measures to various contexts (Wolters & Pintrich, 1998). Participants completing these measures are directed to provide either prospective or retrospective reports related to a specific task. Prospective strategy measures, completed prior to engaging in a task, instruct individuals to identify the strategies that they plan to use while completing the task (e.g., Cromley & Azevedo, 2006). In contrast, retrospective strategy measures occur following a task and instruct participants to report the strategies that they implemented during the given task (Bråten & Strømsø, 2011; Murphy & Alexander, 2002).

The assignment of strategy conditions in experimental research is a methodology used when task variations are believed to elicit different types of strategic processing (e.g., Grodsky & Giambra, 1990; Freed, Marshall, & Phillips, 1998) or when individuals are taught different strategies (or no strategies) depending on their condition (e.g., L. H. Mason, 2004; Spörer, Brunstein, & Kieschke, 2009). Conditions that differ according to task variations are frequently associated with studies examining differences in deep- and surface-processing strategies (Dinsmore & Alexander, 2012). For instance, in a study on reading comprehension strategies, students were instructed when to use the specific strategies of rereading, paraphrasing, and self-explaining (Moss, Schunn, Schneider, McNamara, & VanLehn, 2011). These different types of strategies were verified in terms of self-reported use and then correlated with regions of brain activation. Strategy intervention research often follows the method of assigning some groups of individuals to receive instruction of the use and implementation of certain strategies, while other individuals serve as a control (e.g., Graham, Harris, & Mason, 2005). In these conditions, strategies are frequently taught one at a time or as small clusters of strategies (Spörer et al., 2009). Rather than examining strategy choice or spontaneous implementation, groups are frequently compared on outcome measures to examine the effect of teaching individuals specific strategies on their subsequent task performance (Grodsky & Giambra, 1990; L. H. Mason, 2013).

Real-time processing measures.

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Think-aloud methodologies have been recognized in the strategic-processing literature as a means of capturing in-the-moment processes. These methods originated in expert/novice and problem-solving research (e.g., Ericsson & Simon, 1980; Simon & Chase, 1973) and later expanded to reading (Pressley & Afflerbach, 1995). Unlike the questionnaire measures, think-aloud methods provide a measure of real-time processing, asking participants to talk aloud and report what they are doing as they are completing a task or solving a problem. Processes are then coded according to the types and specific strategies reported. Amid concerns that thinking aloud during a task such as reading changes the process or that individuals are not able to accurately report the strategies they employ (Dunlosky & Hertzog, 2001; M. C. Fox, Ericsson, & Best, 2011; Pressley & Afflerbach, 1995), these methodologies continue to be frequently used within the strategic-processing literature (Afflerbach, 2000; E. Fox, 2009).

Eye tracking provides a complementary of real-time processing (Just & Carpenter, 1980) through the examination of eye movements (i.e., saccades). It is used both independently and concurrently with other measures such as think-alouds and questionnaires (Kaakinen & Hyönä, 2005). As a measure of strategy use, eye-tracking measures have followed a similar historical trajectory as think-alouds, emanating in research on expertise and problem-solving (Abernethy & Russell, 1987; Just & Carpenter, 1976) and later implemented for use in studies of reading (Just & Carpenter, 1980; Rayner, 1978). As a measure of strategic processing, eye tracking provides data about the time spent on specific visual elements, such as the time spent on answer choices to a multiple-choice problem relative to the time spent on the stem (Vakil, Lifshitz, Tzuriel, Weiss, & Arzuoa, 2011; Vigneau, Caissie, & Bors, 2006). During reading tasks, eye tracking provides information about the time spent on individual words, the number of look backs to specific words, and the relative time spent on text compared to other visual elements such as graphs (Ariasi & Mason, 2011; Hannus & Hyönä, 1999; Hyönä, Lorch, & Kaakinen, 2002). From these data, researchers make inferences about the use of processes such as rereading, understanding a word in context, and the identification of anomalous words (Rayner, Chance, Slatterby, & Ashby, 2006). Although eye tracking has existed for more than a century (Rayner, 1998) and was available when Alexander and Judy (1988) published their review, it has been more prominent in strategy research over the past several decades (Rayner, 1998).

In contrast, the use of neuroscientific methodologies in strategic-processing research was virtually nonexistent 25 years ago. While neurophysiological methods have remained limited in both their frequency of use and the degree to which they can capture real-time strategic processing, they have received increasing interest in recent years (e.g., Grabner & De Smedt, 2012; Newman, Willoughby, & Price, 2011). At the current time, most technologies associated with neurophysiological investigations of strategy use present important advantages and disadvantages.

For example, functional magnetic resonance imaging (fMRI) machines collect information about the location and sequence of a participant's brain activation while doing a task—allowing for an unprecedented view of cognition—but, at the same time, limit the types of

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tasks that can be completed by participants. Specifically, while inside an fMRI machine, participants cannot move or speak and therefore are often limited to pressing buttons to select response options during a task (Rosenburg-Lee, Barth, & Menon, 2011) or to reporting of strategies outside the scanner (Cho et al., 2011; Grabner et al., 2009). As such, it is common for fMRI studies of strategic processing to assign several tasks that are presumed to require different types of strategies with patterns of brain activation compared across task type (Moss et al., 2011).

In comparison, methods such as functional near-infrared spectroscopy (fNIRS) and event-related potentials (ERPs) allow a participant to move his or her head and to speak during an experiment, which provides for a much wider range of tasks to be used but comes with their own specific limitations. For example, participants can think aloud or interact with one another while an fNIRS machine records the activation in their cerebral cortex (Dresler et al., 2009; Hatta et al., 2009), but activation in other, further-from-the-skull brain regions cannot be detected. We return to the discussion about these new technologies and their associated methodologies in a later section of this chapter.

Expanding Classes of Strategies

In the years since the publication of Alexander and Judy's (1988) review, much has been learned about the characterization of strategies and strategy use. But, in these intervening years, has the field come to identify new classes or types of strategies that warrant further examination? That is to say, has the field grown not only in the relative depth of what is known, but also in the breadth of strategic processes that are under investigation? In this section, we offer two principal areas of study in which empirical research and accompanying scholarly debate have gained prominence in recent years and in which, in our estimation, scholarly interest is likely to continue to grow: (a) metastrategies and (b) strategies for online learning. We turn to a critical examination of each of these areas of study, explicating these types of strategies and whether they are in fact "new" types of strategies or iterations of previous conceptualizations.

Metastrategies

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Metastrategies are strategies that learners employ to regulate or monitor their learning, understanding, and strategy use. In this way, some of the types of strategies previously described in this chapter, like metacognitive strategies (e.g., Pintrich, 1991) and self-regulatory strategies (e.g., Schunk, 2008), can be considered members of the class of metastrategies. The key difference between metastrategies and other cognitive learning strategies is that metastrategies are used to guide or monitor an individual's use of more specific cognitive strategies. This guiding or monitoring of cognitive strategies by metastrategies can occur across a variety of domains. In a sense, metastrategies are always domain general, although it is likely that individuals are not as adept in applying these strategies in all domains (Schraw, 1998; Veenman et al., 2006). Whenever a learner is strategically regulating personal strategy use, the learner can be said to be employing a metastrategy.

While descriptions of metastrategies themselves (i.e., metacognitive and self-regulatory strategies) have existed in the educational and psychological literature for some time (e.g., Flavell, 1981; Garner, 1987), recent years have seen the class of metastrategies broadened in significant ways. Specifically, the characterization of relational reasoning strategies as metastrategies has opened new opportunities for the study of metastrategies and the academic outcomes to which they are linked. These relational reasoning metastrategies, their utility to learners, and their associated research traditions are now discussed.

Relational reasoning.

Relational reasoning metastrategies are cognitive procedures purposely applied to recognizing or deriving meaningful relations or patterns between and among pieces of information that may otherwise appear unrelated (Alexander & the Disciplined Reading and Learning Research Lab [DRLRL], 2012). The ability to employ these strategies is regarded as fundamental to human cognitive function (Hofstadter, 2001; James, 1890; Sternberg, 1977) and education (e.g., Richland, Zur, & Holyoak, 2007). It has been identified as an important ability associated with early learning (Goswami, 1992), as well as a hallmark of expertise, especially in scientific domains (Dunbar, 1995). Relational reasoning strategies require students to engage in deeper level processing of information than they may otherwise do (Holyoak, 2012) and may counteract the tendency of students to treat information in an isolated or piecemeal way (Braasch & Goldman 2010).

While the bulk of the literature on relational reasoning characterizes the construct as an ability or trait and not as a body of strategies available for students to learn and employ (Krawczyk, 2012), a growing body of work puts strategic processing at the forefront of conceptualizations of relational reasoning (e.g., Alexander & the DRLRL, 2012; Alexander, Dumas, Grossnickle, List, & Firetto, 2016; Richland & McDonough, 2010). This characterization of relational reasoning as a body of metastrategies to be employed has potential to be informative for programs of research that focus on both relational reasoning specifically and metastrategies in general.

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Here, we briefly discuss four particular relational reasoning metastrategies that have been put forward as warranting further systematic investigation (e.g., Alexander & Baggetta, 2014; Alexander et al., 2016). These four metastrategies have been extensively reviewed elsewhere (Alexander & the DRLRL, 2012; Dumas, Alexander, & Grossnickle, 2013). As such, the present discussion is relatively brief. The four relational reasoning metastrategies considered in this chapter are analogy, anomaly, antinomy, and antithesis.

Analogy.

An analogic strategy is a procedure purposely applied to the identification of a relational similarity between two or more concepts, objects, or situations (Bostrom, 2008; Hesse, 1959). The ability to employ analogical reasoning strategies has been shown to be instrumental in knowledge transfer (Gentner, Loewenstein, & Thompson, 2003) and has been empirically connected to learning in multiple broadly conceived academic domains, including mathematics (Alexander, White, & Daugherty, 1997); science (Trey & Khan, 2008); reading (Schiff & Ravid, 2007); and writing (Hollaway, 2007). Analogical thinking and its associated metastrategies may also be essential for the organization of memory and the development of language (Hofstadter & Sander, 2013).

Anomaly.

An anomaly can be any occurrence or object that is abnormal or that digresses or deviates from an established pattern (Ferguson & Sanford, 2008; Trickett, Trafton, & Schunn, 2009). The ability of students to strategically recognize or discern an anomaly has been tapped for over a century on tests of intelligence (Binet, Simon, & Town, 1913). It has also been shown that attending to anomalies is critical for conceptual change in science (Chinn & Malhotra, 2002). Some theorists argue that the process of anomaly resolution facilitates an individual's understanding of his or her implicit assumptions and logical mistakes (Darden, 1995), and that anomaly detection is the principal way in which scientific fields advance (T. S. Kuhn, 1996). Similar to analogical reasoning, students' ability to employ metastrategies associated with attending to and reasoning with anomalies has been empirically linked to academic success in multiple academic domains, including mathematics (Chen et al., 2007); science (Trickett et al., 2009); and reading (Bohan & Sanford, 2008).

Antinomy.

Whenever a student strategically seeks to understand what something is by describing what it is *not*, the student is employing an antinomial metastrategy. This metastrategy can be particularly helpful in paradoxical situations in which a domain of study has come to accept two or more ideas that appear contradictory, when the issue of mutual exclusivity among categories, or the incompatibility of theoretical models is being reasoned with (Chi & Roscoe, 2002; Opfer & Gelman, 2011; Sorensen, 2003). While the empirical study of students' use of antinomial metastrategies remains largely underexplored (Dumas et al., 2012), the identification and consideration of antinomies have led to productive discussion and even discovery in a variety of academic disciplines,

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including educational psychology (Cole & Wertsch, 1996); mathematics (Russell & Lackey, 1973); psycholinguistics (Shaumyan, 2006); reading (Mosenthal, 1988); and intelligence (Gardner, 1995).

Antithesis.

The final relational reasoning metastrategy discussed in this chapter, antithetical reasoning, implies the recognition of a directly oppositional relation between two mental representations (Heit & Nicholson, 2010). The ability to strategically conceptualize antithetical opposition is important for argument and persuasion (D. Kuhn & Udell, 2007) and conceptual change with refutation text (Broughton, Sinatra, & Reynolds, 2010). Some scholars have argued that human language and thought are primarily organized through relations of opposition (De Saussure, 1916/2011; Kjeldergaard & Higa, 1962; Marková, 1987), implying that antithetical metastrategies may have a deep-seated importance in human communication and cognition.

The investigation of each of these four relational reasoning metastrategies—analogy, anomaly, antinomy, and antithesis—shows promise to be a potentially fruitful avenue of inquiry for those interested in students' deeper processing of information and their subsequent building of principled knowledge. Because students' ability to employ these metastrategies may be especially crucial in the modern information-saturated context (Dumas et al., 2013), we believe that the systematic examination of relational reasoning metastrategies—as well as how these metastrategies interface with more specific cognitive strategies and associated individual difference characteristics—will be an important addition to the larger literature on students' strategic and metastrategic behavior.

Strategies for Online Learning

Another class of strategies that has entered into scholarly discourse in recent years is those strategies associated with learning on the Internet, including those involved in learning from hypertext (Afflerbach & Cho, 2010). Hypertext can be defined as written text, pictures, video, or sound connected by electronic links (Rasmussen, 2007). The collection of skills and strategies employed by learners in a hypertext environment has been referred to as “new-literacies” (e.g., Leu et al., 2008). As is implied by the use of this term, many scholars and researchers who either theoretically describe or empirically investigate this class of strategies conceptualize them as wholly novel and arising from the need for readers to interact with new technologies and texts (e.g., Coiro, 2003).

However, while neither the observation that the sheer quantity of information available almost instantly via the Internet may present challenges and opportunities to students nor the assertion that readers' strategies for dealing with this information and potentially deriving knowledge from it should be of interest to researchers are debated, significant questions remain about the characterization and classification of hypertext reading strategies. Specifically, while some scholars characterize hypertext reading strategies to

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be entirely new and qualitatively distinct from strategies that could be employed by students reading traditional texts (e.g., Leu et al., 2008), others in the field have argued that, while there may be differences in the relative quantity and frequency of strategies that are employed when reading from hypertext versus traditional text, the strategies utilized when reading hypertext are basically iterations of strategies that have long been employed by readers of traditional text (e.g., Alexander and the DRLRL, 2012).

For example, Afflerbach and Cho (2009) argued that certain strategies associated with hypertext reading “appeared to have no counterpart in traditional reading” (p. 217). They specifically pointed to strategies that had to do with “realizing and constructing potential texts to read” (p. 217). Afflerbach and Cho (2009) offered several specific examples of these strategies, including searching for relevant websites or information retrieval systems to access and overview possible target information; reducing the range of possible information to be encountered by generating keywords related to the topic and focus of a particular task; and scrutinizing Internet hypertextual links to anticipate and judge the usefulness and significance of the information before accessing it, based on specific reading goals; among others. However, these specific strategies could also be conceptualized as iterations of strategies that have long been used in traditional learning. Internet search strategies can be conceived as having an analog in traditional text searches.

For instance, for as long as there have been collections of informational text (i.e., libraries), readers have needed to search various catalogs or organizational systems for information. These searches required the positing of keywords and the scrutiny of content summaries of various available texts before choosing one or more to read more deeply. While the technological platforms for these searches may change, their strategic function remains the same—to access the desired information in an efficient, effective way. In other words, new literacy strategies may not be Internet or hypertext specific but may have been employed in different technological contexts by generations of strategic learners.

This debate is important not only for the theoretical conceptualization of hypertext reading strategies but also for the study and teaching of cognitive learning strategies at large. This is because the two competing conceptualizations of hypertext reading strategies make divergent implications for the fields of education and psychology. For example, if the strategies utilized by readers in a hypertext environment are wholly new and qualitatively different from those used when reading traditional text, then high-leverage strategies and metastrategies that have been perennially linked to student success—like the relational reasoning strategies described previously in this chapter—may not be able to be expected to translate to a hypertext environment. This would necessitate the teaching of hypertext-specific skills and strategies to students if they were to succeed in the twenty-first-century context. However, if, as we argue, the same skills and strategies that have undergirded success with traditional text (e.g., connecting to prior knowledge, making inferences about information not explicitly stated, questioning the author, relating aspects of the text to one another) will continue to be advantageously

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employed by learners in hypertext environments, then these historically important cognitive strategies will not diminish in importance but will iterate to new forms of use as the technological platform for reading changes. Importantly, the iterative conceptualization of strategies allows the cognitive skills and strategies that are important today to continue to be important into the future, whereas the new-literacy conceptualization requires the salient skills and strategies to be ever changing, depending on the technological context of the times.

Of course, we are not arguing that the specific instruction of computer or Internet skills is unimportant or should not be included in school curricula. Because these technologies are so ubiquitous today and a certain aptitude with them is likely required for any learner to succeed either in school or in the workplace, their use is certainly an important topic for instruction. In fact, because of the necessity of interacting with this technology, it is no surprise that investigations like that of Coiro (2011) have found significant effects of students' online research ability to their hypertext reading comprehension. However, we do argue that the advent of hypertext does not, as Afflerbach and Cho (2010) have stated, represent a "change in the architecture of reading" (p. 217), but rather it signifies a shift in the platform on which the act of reading is performed. This shift is likely to change the way cognitive strategies for reading manifest themselves and potentially alter the relative importance of certain strategies and skills, but, in our estimation, it will not fundamentally change the act of reading or the cognitive skills, strategies, or metastrategies required to be a successful reader now or in the future.

Strategies in the Classroom

In this chapter, we endeavor to show that students who are engaged in strategic behavior learn and perform better than students who are not strategic, and that strategic processing develops over time (Bråten & Strømsø, 2003; Murphy & Alexander, 2002) and across domains and contexts (Alexander & Judy, 1988; Butler, 1995; Murphy & Alexander, 2002). Being strategic requires that students have a repertoire of strategies and that they understand when and how to use them (Slater & Horstman, 2002). Intervention research over the past 25 years has indicated that teachers have the potential to help their students learn strategies (De La Paz, 2005; Jitendra et al., 2009; Keys, Hand, Prain, & Collins, 1999; Klinger, Vaughn, & Schumm, 1998; L. H. Mason, Snyder, Sukhram, & Kedem, 2006). Yet, what challenges do teachers and students face in order for students to be good strategic processors, and what recommendations can be put forward to alleviate some of these challenges?

What Are the Barriers to Being Strategic in Classrooms?

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Within the classroom setting, teachers and students face barriers that prevent students from being strategic. Alexander and Judy (1988) stated, “It could be that schools perpetuate the accumulation of inert knowledge or reward routinized nonstrategic behavior” (p. 398). While they did not elaborate much on this point, others have noticed a similar pattern. Writing around the same time as Alexander and Judy, Pressley, Goodchild, and colleagues (1989) addressed challenges to strategy instruction, emphasizing the importance of knowing when and where to use strategies. They also provided a list of suggestions for practitioners to address in order to aid students in their strategic processing, including recommendations to teach across-domain strategies and to motivate students to use the strategies they are taught.

Building on the precedent for strategy instruction research set by Pressley and others (Duffy & Roehler, 1989; Swanson, 1989), Ruth Garner, in her 1990 article “When Children and Adults Do Not Use Learning Strategies: Toward a Theory of Settings,” explained why strategies are often not used within academic contexts. Garner put forward five reasons: poor cognitive monitoring, the use of primitive routines, knowledge deficiencies, goals that do not support strategic processing, and minimal transfer to novel situations. For students and teachers in the twenty-first century, these barriers may manifest differently, but we argue that these reasons continue to challenge strategic processing.

Poor cognitive monitoring.

First, Garner (1990) noted that strategic processing is hindered because students exhibit poor cognitive monitoring. She suggested that, “If children and adults do not notice that they are not learning, they are unlikely to seek a strategic remedy” (p. 518). As is noted in research on metacognition, metacognitive strategies function to guide cognitive strategies, and the two are inextricably intertwined (Flavell, 1979). If inefficient or ineffective metacognitive strategies are employed, then students may falsely believe that they understand the task when they may actually only grasp a portion of it (Garner, 1990).

In our estimation, systematic investigations aimed at improving students’ cognitive monitoring will not diminish in importance as the twenty-first century progresses. Rather, for today’s students, cognitive monitoring becomes even more essential. It takes place in a world where information is readily available in both print and digital formats (Bohn & Short, 2009) and where using multiple devices and completing multiple tasks simultaneously is considered the norm (Alexander & the DRLRL, 2012; Rideout, Foehr, & Roberts, 2010). Poor cognitive monitoring often results in surface-level rather than deep processing of vast information, preventing students from realizing that there is more to be learned from the content (Garner, 1990), a particular concern for learners in the information age. Given these certain aspects of the modern technological context, the ability of students to monitor their own learning—as well as empirical investigations of that ability—will likely remain significant.

Primitive routines.

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Second, Garner (1990) stated that primitive routines help students produce a product, but she noted that they “inhibit the use of learning-enhancing strategies” (p. 519). For example, some summarizing routines encourage students to copy texts verbatim instead of requiring the reader to think deeply about the text and consider what information is important to keep or delete in the summary (Garner, 1990). After using this summarizing routine, an effortless product is produced, but the routine prevents students from using strategies that would help them think more deeply about the text. These dysfunctional routines continue to hinder students’ strategic processing today. Many twenty-first-century students acquire information by conducting Internet searches. Students often engage in the primitive routines of selecting the first source in a search, reading the title and the first couple of sentences, and quickly skimming the article, attempting to get the gist of the text (Guinee, Eagleton, & Hall, 2003). However, students lack the ability to strategically synthesize information from multiple sources or evaluate the sources they encounter (Metzger, Flanigan, & Zwarun, 2003; Walraven, Brand-Gruwel, & Boshuizen, 2009). Students’ maladaptive routines help them access the information online but do not help them make sense of the content or evaluate sources effectively (Bråten, Strømsø, & Britt, 2009; List, Grossnickle, & Alexander, 2012).

Meager knowledge base.

Third, Garner (1990) put knowledge deficiencies forward as another cause of students failing to use strategies. She provided the following two examples of students for whom a lack of knowledge hindered their ability to be strategic:

Think for a moment about a high school student who knows it is wise to prepare for an essay exam and a multiple choice exam in different ways; however, he has not been given any information by his instructor about the format of the next day’s test. Or, think about a third grader who knows she should spend more time on difficult words than on easy words in preparing for a spelling test, but she does not know how much time is available. (p. 520)

In these situations, the absence of knowledge is due to external factors, such as information from a teacher or a parent, which would better help the students strategically study for their assessments. However, having a meager knowledge base also encompasses a lack of content-specific background knowledge (Kaplan & Murphy, 2000). For example, a student cannot analyze the causes of the Civil War if the student does not understand the economic differences between the North and the South during the early 1860s. Additionally, a student cannot successfully learn long division if the student struggles with addition and subtraction.

Because students today are surrounded by more information from multiple sources, they need to build expansive knowledge bases and use their background as a launch pad for strategic processing. However, twenty-first-century learners often focus on information management rather than deep thinking and knowledge building (Alexander & the DRLRL, 2012). They gather bits of material when they believe it is relevant to their current situation but do not seek to retain that information or develop it as a part of their

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knowledge base (Alexander & DRLRL, 2012; Alexander & Kulikowich, 1991; Payne & Reader, 2006). In a classroom setting, students spend their time skimming the surface of texts or memorizing facts for a test without seeking understanding of the content (Cates, Burns, & Joseph, 2010; Schwartz, Sadler, Sonnert, & Tai, 2009). Individuals are overloaded with information, yet fail to exhibit strategic behavior that would help them make sense of the information and engage in critical thinking.

Classroom goals.

Fourth, Garner (1990) explained that classroom goals do not always support strategy use. Specifically, she indicated that classrooms tend to support either performance or mastery goal orientations, a finding that has been supported in recent literature (Murayama & Elliot, 2009; Patrick, Kaplan, & Ryan, 2011). If students maintain a performance goal orientation (i.e., are motivated by demonstrating competence on outcomes relative to others; Ames & Archer, 1988), then they may not exhibit strategic activity, which requires effort and investment in the learning process (Pressley, Goodchild, et al., 1989). Within the twenty-first-century context, classroom goals focus predominantly on acquiring high grades and doing well on standardized assessments (Abrams, Pedulla, & Madaus, 2003; Guzenhauser, 2003; McNeil, 2000). Curricula focus on a breadth rather than a depth of knowledge; therefore, teachers reward students for simply regurgitating surface-level information. Students are not assessed on their strategic processing but instead are assessed on their ability to restate information or follow a heuristic when told to do so.

Additionally, standardized tests are increasingly important as educational systems seek to link teacher pay to student test performance (Springer & Gardner, 2010; Woessmann, 2011). As a result, teachers focus on techniques that will help students excel on a test instead of motivating students to learn complex content and develop thoughtful ideas about domain knowledge. Classroom conversations revolve around the ways to eliminate incorrect answer choices rather than analyzing arguments or developing opinions. The goal of school moves away from equipping students with essential knowledge and the ability to think strategically in new situations and instead emphasizes passing high-stakes assessments that require more rote processes than strategic processing (Black & Wiliam, 1998). If teachers do not reward students for high-level thinking throughout the year, then they will not have an incentive to use strategies in the classroom.

Minimal transfer.

Finally, Garner (1990) explained that students exhibit minimal transfer of strategic activity to new situations. She noted:

Teachers and researchers alike have observed an absence of generality of strategy use among students. It is usually the intention of teachers to instruct students in how to use a particular routine to solve a set of related problems; yet it is usually the practice of students to apply the routine, if at all, to only the subset in which

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the instruction was originally provided. In other words, the strategy is welded to the original instructional context. (p. 522)

Students have trouble applying their knowledge when problem solving, encountering a new text or new content, or bootstrapping their prior knowledge to help themselves think relationally between various concepts (Alexander & the DRLRL, 2012). At the same time, transfer is only helpful when strategic knowledge is present. Students cannot transfer their strategic knowledge if they are not learning how to be strategic in the first place.

Within the twenty-first-century context, students continue to struggle when encountering new situations. For example, a student might learn a questioning strategy to help comprehend the fictional novel *Bridge to Terabithia* by Katherine Paterson. However, the student may not realize that she can adjust this same questioning strategy and apply it when reading an article about global warming, solving a mathematics word problem, watching a video clip of Martin Luther King Jr.'s *I Have a Dream* speech, searching the Internet for information about current politicians, or engaging in social media websites. Today, students confront a variety of texts and problems, yet they often do not think strategically when encountering novel contexts.

What Do Learning Settings That Promote Strategies Look Like?

Despite these barriers to being strategic in a classroom, there are situations in which teachers and students can foster strategic thinking. We suggest that strategic thinking is promoted when (a) teachers and students understand what it means to be strategic, (b) teachers and students know how to apply strategies appropriate to the context, and (c) learner differences are considered.

Teachers and students understand what it means to be strategic.

For strategy instruction to be effective, it is first necessary for teachers and students to understand what it means to be strategic. As mentioned earlier, this is not always the case. Researchers, practitioners, and students often lack clarity regarding the differences between strategies and skills, as well as differences between various types of strategies (Afflerbach et al., 2008). Definitional issues have important instructional implications. Teachers often mistake routinized skills for purposeful strategy use and subsequently focus on teaching skills instead of strategies.

For students, this confusion may result in part from the frequent teaching of a series of steps to follow when attempting a particular task (Bailin, Case, Coombs, & Daniels, 1999). For example, teachers might require students to follow the “sandwich model” of writing an essay, a model that provides students with a five-paragraph formula for writing an essay with an introduction, conclusion, and three supporting points. The assumption is that, in the end, students will compose a well-structured essay; however, such a formulaic

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structure is not well suited for many writing tasks. While step-by-step approaches work in certain situations, they do not include the totality of what it means to be strategic.

Additionally, understanding what it means to be strategic implies that strategies can be employed across a variety of situations. As such, strategy instruction in the classroom should focus on teaching for transfer so that students can apply the strategies in a variety of contexts. For example, teachers can broaden the categories they use when asking students to group mathematics problems (Fuchs et al., 2003). Teachers should also provide opportunities for students to strategically solve novel problems (Fuchs et al., 2003), and students should take advantage of their teachers' prompts. Encouraging students to build a repertoire of strategies, to personalize and take ownership of those strategies, and to apply them adaptively when encountering novel situations aligns with the definition of strategies and supports students' efforts to be strategic.

Teachers and students know how to apply strategies appropriate to the context.

Practitioners and students should consider the context in which strategic learning occurs. Garner (1990) explained, "One thing that we already know about strategy use is that it is embedded. It does not occur in a vacuum. When context varies, the nature of strategic activity often varies as well" (p. 523). In order for students to be able to pick strategies appropriate to the context, teachers and students should bundle strategies so that students do not use too many at once or one in isolation. Using a single strategy often does not provide students with the desired result of comprehending a text or discovering the solution to a difficult problem. Rather, a few strategies should be used in conjunction so that students can determine which strategies work best under what circumstances (Palincsar & Brown, 1984).

The focus on bundling strategies emphasizes that not only may more than one strategy be appropriate in a given situation, but also helps students make sense of the strategies they are taught instead of becoming overwhelmed by too many options. For example, Palincsar and Brown (1984) taught summarizing, questioning, clarifying, and predicting strategies to seventh-grade students who were poor readers. When the strategies were taught together, learners' reading comprehension improved, as demonstrated by both standardized and classroom comprehension assessments, and gains were maintained over time with the ability to utilize the strategies in new situations. As Palincsar and Brown (1984) suggested, students sometimes need to use multiple strategies simultaneously. However, students do not necessarily need all strategies (Pressley & Afflerbach, 1995), so it is important to support students in determining which strategies help lead them to success.

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Learner differences are considered.

Finally, strategy instruction would benefit from taking individual differences into account. The major contribution of Alexander and Judy's (1988) article was to examine the interrelation between strategic processing and domain knowledge. Strategies do not act in isolation, rather they are initiated, enacted, and monitored by a learner who comes to any given situation with individual variation in knowledge as well as other factors, such as epistemic beliefs and motivation (Richter & Schmid, 2010; Shen & Chen, 2006; Strømsø & Bråten, 2010). While Alexander and Judy's focus was on domain knowledge and strategic processing, the consideration of interactions between strategies and individual differences more broadly is evident in the field today and provides a means by which learning settings can support strategy use.

With regard to the interaction of knowledge and strategies, Alexander and Judy (1988) hypothesized domain knowledge is necessary for strategic processing, and that strategic processing is necessary for the use and acquisition of domain knowledge. Given the intervening years of empirical research supporting this hypothesis, learning environments should take students' domain knowledge into account as a means of supporting strategic processing. Specifically, providing students with requisite domain knowledge or opportunities to access their prior knowledge before engaging with tasks requiring strategic processing is critical. However, this is often not the case. As schools and standards move away from goals emphasizing the acquisition of domain knowledge toward goals emphasizing higher order thinking in the absence of knowledge goals (CCSS, 2010), the consideration of learner differences in domain knowledge and the importance of this reciprocal relation is often deemphasized, despite evidence of the central interaction.

In addition to domain knowledge, students' beliefs about the nature of knowledge (i.e., epistemic beliefs) have been examined more recently in relation to strategic processing (Bråten & Strømsø, 2006; Murphy & Mason, 2006; Sinatra & Pintrich, 2002). Understanding why an individual is successful or unsuccessful in implementing strategies is linked in important ways to a student's epistemic beliefs, given that certain configurations of beliefs tend to be associated with more successful strategy use (Muis & Franco, 2009; Strømsø & Bråten, 2010). Consideration of a student's epistemic beliefs allows for an understanding of the knowledge related to the task from the lens of the student. If the student perceives knowledge about history to be simple and handed down from authority, this understanding would provide those within the learning setting the opportunity to scaffold the tasks provided for that student to guide the his or her understanding to include beliefs about the complexity of knowledge and the need for justification.

Alexander and Judy (1988) briefly foreshadowed the recent surge of interest in motivation and strategic processing, noting that "motivational and social-contextual factors may impact the acquisition and utilization of domain-specific and strategic knowledge" (p. 395). Although their review identified motivational factors as cursory variables in the studies they reviewed, interest in the motivation-strategy relation has expanded widely in

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these years since their work (Alexander & Murphy, 1999; Linnenbrink-Garcia & Pekrun, 2011; McCrudden, Magliano, & Schraw, 2010). In fact, it is difficult to consider the effortful, purposeful manner that is required to be strategic without a discussion of motivation (Alexander et al., 1998). Consequently, concern for how individual differences in motivation can be leveraged to support strategic processing needs to be considered in relation to the setting in which learning occurs and to the responsibility of learners within that setting to regulate and support their motivations. Although motivational research in areas such as need for cognition (Cacioppo & Petty, 1982) and trait curiosity (Litman, 2010; Silvia & Sanders, 2010) suggest that motivations related to learning are dispositions to engage in the effortful behaviors required of strategic processing, there is substantial evidence that characteristics of academic tasks and the environment can be orchestrated to support motivational differences in a way that leads to increased strategic processing (Hulleman & Harackiewicz, 2009; van Dinther, Dochy, & Segers, 2011).

Into the Future: What the Next 25 Years of Strategy Research Holds

What we have sought to accomplish to this point in our examination of cognitive learning strategies was to revisit the groundbreaking article by Alexander and Judy (1988) and to ask whether the issues and concerns these researchers articulated years ago are as problematic or as pertinent today. This was the retrospective mission we set for ourselves. Now, in this closing segment of the chapter we look ahead to the next 25 years. In effect, based on where we have come, what would we hold to be the future directions for theory and research on cognitive learning strategies? In preparing for this prospective look at cognitive learning strategies, we have decided to be bold and venturesome in our prognostications, building on the trends that have begun to emerge within the contemporary literature and referenced herein. Specifically, we address four topics that we view as critical for continued progress in the area of cognitive learning strategies and that we hope will be richly explored in the years to come: (a) developmental and longitudinal investigations of strategy use; (b) neurophysiological indicators of strategic thinking and learning; (c) innovative measures and data-analytic procedures for the study of strategic processing in situ; and (d) academic environments constructed to promote strategic thinking and performance for all learners.

Developmental and Longitudinal Studies

While we have come to learn so much about strategies and the role they play in academic development over the years since Alexander and Judy's (1988) work, as well as about the interplay between strategies, knowledge, epistemic beliefs, and motivations/emotions, so much of that understanding has come from cross-sectional or short-term studies (Bråten

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& Strømsø, 2011; Murphy & Alexander, 2002). For instance, comparing the strategic behavior of elementary versus middle or high school students (e.g., Meneghetti & De Beni, 2010); investigating the strategic thinking of novices in a domain versus those with demonstrated expertise (e.g., Ericsson & Simon, 1980); and establishing the significant effect of a given classroom intervention on students' learning in science, mathematics, or history (De La Paz, 2005; Jitendra et al., 2009) have collectively illuminated the power of strategic processing to subsequent learning and performance. Yet, even the impressive library of strategy-related research that now exists cannot serve to answer fundamental questions about how strategic processing—in conjunction with knowledge or motivation—should systematically change over the course of one's academic development.

Several years after the publication of Alexander and Judy's 1988 article, Alexander (1997) theorized how such changes might unfold over time within academic domains. That is, what would the interplay of strategic processing, knowledge, and interest look like as individuals progress from the initial stage of learning (i.e., acclimation) into competence or even expertise? That theorizing took the form of the MDL (Alexander, 1997, 2003), mentioned previously. However, while numerous studies of the MDL in a variety of academic domains have been undertaken (e.g., Kultural-Konak, Konak, Esparragosa, & Kremer, 2013; Langan & Athanasou, 2005; Schrader, Lawless, & Mayall, 2008), there remains no long-term, longitudinal examination of its viability. Yet, it is precisely such developmental and longitudinal investigations that are requisite to push understanding of the relation between cognitive strategies and academic development to the next level. Thus, should we return to this topic of cognitive strategies in 2038 or thereafter, we would hope to see a wealth of longitudinal investigations populating the literature and a more extensive understanding how strategic processing, in fact, develops over the life span.

Neurophysiological Traces of Strategic Thinking

Even though new technologies for the assessment of strategic processing are far more available today than in 1988 when Alexander and Judy wrote their article, the field of educational psychology still relies predominantly on self-report measures to chronicle strategic processing, be those measures real time, prospective, or retrospective in nature. However, if the increased presence of technologies like eye tracking or neurocognitive devices such as fMRI, ERP, and fNIRS, as well as their associated methodologies, continue to gain popularity as our previous discussion suggested, there is good reason to expect that the inclusion of neurophysiological indicators of strategic processing will be much more commonplace.

Thus, in the literature on strategies published 25 years from now, researchers might be expected to do more than merely ask students to report what they strategically are planning to do (prospective), are doing (real time), or did (retrospective). Whenever possible, researchers of the future might be called on to incorporate neurological or physiological data that can corroborate such self-reports. One can even envision a time

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and place where self-report data come to support or extend findings about strategic processing secured principally through neurophysiological channels. Of course, whether this prognostication comes to fruition will depend to a great extent on the availability and affordability of neurophysiological devices or the degree to which the knowledge needed to acquire and interpret data from such devices becomes de rigueur for educational psychology researchers.

Advances in Measurement and Statistics

Invariably, advancements in what can be learned in any complex arena are inextricably tied to the availability of measures and statistical procedures that allow for the gathering and interpreting of relevant data. So it has been for the study of cognitive learning strategies. In 1988, when Alexander and Judy first published their article, measures of strategic processing that were considered psychometrically sound were few and far between. It remains true today that measures of strategic processing are relatively limited in number and the psychometric properties of such measurement tools that often rely so heavily on self-report data or data pertaining to generic strategy use have increasingly been called in question (e.g., Samuelstuen & Bråten, 2007). Thus, for strategy research to reach the next level in the quarter century to come, it seems imperative that processing measures not only meaningfully reflect the domains or tasks at hand but also demonstrate that the resulting data meet acceptable psychometric standards.

Similarly, such statistical procedures as structural equation modeling, multidimensional item-response theory (IRT), or mixture modeling were not commonly applied to the study of strategic processing—or any other educational construct for that matter—about 30 years ago. Those with the basic knowledge of these approaches were relatively few in number, and existing computer programs did not permit such procedures to be easily or quickly executed. Even if the programs did exist, computers of the time were simply not up to the task. Today, it is relatively easy to run many sophisticated statistical programs that were not even on the radar all those years ago, and many of today's educational psychologists possess knowledge of a range of statistical procedures that would have seemed alien to their predecessors. Of course, how well they understand or appropriately apply those sophisticated statistical procedures is another matter entirely.

Nonetheless, continuing our optimistic stance, we would hope that existing statistical tools continue to be refined and augmented so that conceptual and operational advancements in strategic processing and the multidimensional and dynamic natures of strategies can be duly assessed, modeled, and analyzed.

Orchestration of Supportive Environments for All Learners

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Since the 1970s, there has been the acknowledgment that strategies matter to human learning and development (e.g., Flavell, 1981; Garner, 1987) and that learners cannot be expected to demonstrate effective and efficient strategy use on a routine basis (Garner, 1990). Thus, the need to remove barriers to strategic processing for students and to forge academic settings that promote strategic thinking and learning has continued unabated, as we strongly argued. Although the interest in cognitive strategies within the educational psychology literature has waxed and waned to a certain degree over the past 30 years, strategies will likely always have a place within the research community. Yet, much of the focus on enhancing strategic processing within educational settings over the past decades has targeted those students with special needs or identified learning difficulties (e.g., Fuchs et al., 2003). In effect, researchers engaged in classroom-based intervention in the realm of strategic processing have seen struggling learners and their teachers as their target audience.

Looking forward to the year 2038, however, we would hope that research in strategic processing and academic development manifests a more encompassing view of the merits of devising facilitative learning environments. Armed with the knowledge garnered from developmental and longitudinal studies of strategic processing, prepared to gather both self-report and neurophysiological data, and informed by more precise measures and more sophisticated data-analytic tools, researchers over the next quarter century should seek to construct environments that meet the needs of diverse populations. Such settings must, by necessity, be flexible to the range of tasks and texts that learners encounter and sensitive to the individual differences of learners, including their own intentions and motivations for task engagement.

In essence, it is hoped the days of one-size-fits-all strategic interventions will have passed into oblivion, giving way to more personalized and adaptable academic environments that grow and change as the needs of learners grow and change. Gone, as well, should be any notion that explicit attention to strategies is solely the purview of problem or struggling learners or that strategies need to be taught in isolation as an added component to existing curriculum. In the ideal academic setting, we are envisioning for 2043, cognitive learning strategies will be considered essential and integral aspects of the normal course of instruction for all subject matter domains and threaded naturally throughout that instruction whenever and wherever as the opportunity or need arises.

Concluding Thoughts

As stated at the outset, it was our decision to structure this survey of cognitive learning strategies around Alexander and Judy's (1988) now-classic review, which sought to explore the interplay of strategic processing and domain knowledge—two lines of inquiry that had progressed rather independently until that point. What intrigued us about that article were not only the principles about the relations between strategies and knowledge these authors derived from the literature but also the quandaries that remained for them and the field at that time. Across the years since the publication of that article, it is surprising how certain principles those authors forwarded are now simply accepted as “fact.” For instance, educational psychology researchers routinely acknowledge that the strategic behavior that should be evidenced in the performance of cognitive tasks will vary as a consequence of the knowledge learners possess about the domain and topic at hand. Likewise, the growing literature on conceptual change serves as substantial evidence to Alexander and Judy's (1988) claim that “inaccurate or incomplete domain knowledge may inhibit or interfere with learning” (p. 386).

Just as surprising, areas of inquiry that were only beginning to appear on the horizon with regard to strategies have certainly come into full view over the decades. Nowhere is this more apparent than in the arena of motivation. For instance, far greater understanding of the complex and intricate relation between learners' strategic behaviors and their motivations (and emotions) has been achieved. The goals students' espouse for their academic engagement (Ames & Archer, 1988; Pintrich, 2000), the value or utility that they perceive for the task at hand (Wigfield & Eccles, 2000), and their interest in or curiosity about the topics under study (Ainley & Ainley, 2011; Hidi & Renninger, 2006) have all been shown to matter in the strategies students' report or demonstrate and in the learning outcomes that result.

Yet, for all that the field of educational psychology has come to understand about strategies over the past quarter century and more, many of the quandaries expressed by Alexander and Judy and others (e.g., Alexander et al., 1998; Garner, 1987, 1990) remain. And, even more have come to light (e.g., Afflerbach & Cho, 2008, 2010; Afflerbach et al., 2008). For instance, there is still limited understanding of how individuals' strategic behaviors are influenced by their beliefs about knowledge and knowing (i.e., epistemic beliefs; Alexander & the DRLRL, 2012; King & Kitchener, 2002). There are also lingering questions about whether there are certain dispositional factors, such as a need for cognition or persistence (Bråten, Ferguson, Anmarkrud, & Strømsø, 2013; Cacioppo & Petty, 1982; Petty, Briñol, Loersch, & McCaslin, 2009), or particular cognitive abilities such as rational or relational thinking (Alexander & Baggetta, 2014; Stanovich, 2009; Stanovich, West, & Toplak, 2012) that increase the likelihood of effective strategic behavior. Moreover, despite the appreciation that contextual forces play a role in strategic processing, there is still much to learn about how the immediate environment

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shapes strategic behaviors or how the technological advancements and information saturation that mark contemporary societies will alter the nature of strategic processing (Alexander & the DRLRL, 2012; Leu et al., 2008; Mayer, 2011).

Without question, there is much to be understood about cognitive learning strategies in the next quarter century, not the least of which is how to translate what is already known into academic environments that benefit all learners. We are encouraged by the burgeoning interest in higher order or deeper processing strategies that suggests continued focus on and investment in cognitive strategies within academic settings (e.g., Dinsmore & Alexander, 2012; Schraw & Robinson, 2011). How the testing culture within postindustrial societies or current initiatives such as the CCSS (2010) play into this situation, we cannot begin to say. Still, given the struggles that have already been witnessed over the past 30 years in building academic settings that not only encourage but also nurture efficient and effective strategic processing (Garner, 1990), we remain cautiously optimistic, a characteristic also evident in the concluding paragraph of Alexander and Judy's (1988) review:

The immediate benefit of this review, we hope, is a better grasp of the cognitive research addressing the linkages between domain-specific and strategic knowledge. Perhaps the conclusions that we have reached, the concerns that we have expressed, and the suggestions that we have offered will give rise to research on the interaction of domain-specific and strategic knowledge in the near future. (p. 399)

We bring this retrospective and prospective review of cognitive learning strategies to a close in much the same way. It is our fervent "hope" that the overview we have offered herein and the recommendations for the future we have provided will prove informative and thought provoking to those who will carry the study of cognitive learning strategies forward. We may have glimpsed the past, but the future of this research truly remains to be written.

References

- Abernethy, B., & Russell, D. G. (1987). The relationship between expertise and visual search strategy in a racquet sport. *Human Movement Science*, 6, 283–319. doi: 10.1016/0167-9457(87)90001-7
- Abrams, L. M., Pedulla, J. J., & Madaus, G. F. (2003). Views from the classroom: Teachers' opinions of statewide testing programs. *Theory Into Practice*, 42, 18–29. doi:10.1207/s15430421tip4201_4
- Afflerbach, P. (1990). The influence of prior knowledge on expert readers' main idea construction strategies. *Reading Research Quarterly*, 25, 31–46.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

- Afflerbach, P. (2000). Verbal reports and protocol analysis. In M. L. Kamil, P. B. Mosenthal, P. D. Pearson, & R. Barr (Eds.), *Handbook of reading research* (Vol. 3, pp. 163–179). Mahwah, NJ: Erlbaum.
- Afflerbach, P., & Cho, B. Y. (2008). Identifying and describing constructively responsive comprehension strategies in new and traditional forms of reading. In S. E. Israel & G. G. Duffy (Eds.), *Handbook of research on reading comprehension* (pp. 69–90). New York: Routledge.
- Afflerbach, P., & Cho, B. Y. (2010). Determining and describing reading strategies: Internet and traditional forms of reading. In W. Schneider & H. Waters (Eds.), *Metacognition, strategy use, and instruction* (pp. 201–225). New York: Guilford.
- Afflerbach, P., Pearson, P. D., & Paris, S. G. (2008). Clarifying differences between reading skills and reading strategies. *The Reading Teacher*, 61(5), 364–373.
- Pressley, M., & Afflerbach, P. (1995). *Verbal protocols of reading: The nature of constructively responsive reading*. New York: Routledge.
- Afflerbach, P., & VanSledright, B. (2001). Hath! Doth! What? Middle graders reading innovative history text. *Journal of Adolescent & Adult Literacy*, 44, 696–707.
- Aharony, N. (2006). The use of deep and surface learning strategies among students learning English as a foreign language in an Internet environment. *British Journal of Educational Psychology*, 76, 851–866.
- Ainley, M., & Ainley, J. (2011). Student engagement with science in early adolescence: The contribution of enjoyment to students' continuing interest in learning about science. *Contemporary Educational Psychology*, 36(1), 4–12.
- Alexander, P. A. (1997). Mapping the multidimensional nature of domain learning: The interplay of cognitive, motivational, and strategic forces. In M. L. Maehr & P. R. Pintrich (Eds.), *Advances in motivation and achievement* (Vol. 10, pp. 213–250). Greenwich, CT: JAI Press.
- Alexander, P. A. (2003). The development of expertise: The journey from acclimation to proficiency. *Educational Researcher*, 32, 10–14. doi:10.3102/0013189X032008010
- Alexander, P. A. (2006). *Psychology in learning and instruction*. Upper Saddle River, NJ: Pearson.
- Alexander, P. A., & Baggetta, P. (2014). Percept-concept coupling and human error. In D. N. Rapp & J. L. G. Braasch (Eds.), *Processing inaccurate information: Theoretical and applied perspectives from cognitive science and the educational sciences* (pp. 297–327). Boston: MIT Press.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Alexander, P. A., Dinsmore, D. L., Fox, E., Grossnickle, E. M., Loughlin, S. M., Maggioni, L., . . . Winters, F. I. (2011). Higher-order thinking and knowledge: Domain-general and domain-specific trends and future directions. In G. Schraw & D. Robinson (Eds.), *Assessment of higher order thinking skills* (pp. 47–88). Charlotte, NC: Information Age.

Alexander, P. A., & the Disciplined Reading and Learning Research Laboratory (DRLRL). (2012). Reading into the future: Competence for the 21st century. *Educational Psychologist*, 47(4), 1–22. doi:10.1080/00461520.2012.722511

Alexander, P. A., Dumas, D., Grossnickle, E. M., List, A., & Firetto, C. M. (2016). Measuring relational reasoning. *Journal of Experimental Education*, 84, 119–151

Alexander, P. A., Graham, S., & Harris, K. (1998). A perspective on strategy research: Progress and prospects [Special issue]. *Educational Psychology Review*, 10, 129–154.

Alexander, P. A., & Judy, J. E. (1988). The interaction of domain-specific and strategic knowledge in academic performance. *Review of Educational Research*, 58, 375–404.

Alexander, P. A., & Kulikowich, J. M. (1991). Domain knowledge and analogic reasoning ability as predictors of expository text comprehension. *Journal of Literacy Research*, 23(2), 165–190.

Alexander, P. A., & Murphy, P. K. (1999). Nurturing the seeds of transfer: A domain-specific perspective. *International Journal of Educational Research*, 31, 561–576.

Alexander, P. A., Schallert, D. L., & Hare, V. C. (1991). Coming to terms: How researchers in learning and literacy talk about knowledge. *Review of Educational Research*, 61, 315–343.

Alexander, P. A., Sperl, C. T., Buehl, M. M., Fives, H., & Chiu, S. (2004). Modeling domain learning: Profiles from the field of special education. *Journal of Educational Psychology*, 96, 545–557.

Alexander, P. A., White, C. S., & Daugherty, M. (1997). Analogical reasoning and early mathematics learning. In L. D. English (Ed.), *Mathematical reasoning: Analogies, metaphors, and images: Studies in mathematical thinking and learning* (pp. 117–147). Mahwah, NJ: Erlbaum.

Alfassi, M. (2008). Reading to learn: Effects of combined strategy instruction on high school students. *The Journal of Educational Research*, 97, 171–184. doi:10.3200/JOER.97.4.171-185

Ally, B. A., Gold, C. A., & Budson, A. E. (2009). An evaluation of recollection and familiarity in Alzheimer's disease and mild cognitive impairment using receiver operating characteristics. *Brain and Cognition*, 69, 504–513. doi:10.1016/j.bandc.2008.11.003.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Ames, C., & Archer, J. (1988). Achievement goals in the classroom: Students' learning strategies and motivation processes. *Journal of Educational Psychology*, 80, 260–267.

Anmarkrud, Ø., Bråten I., & Strømsø, H. I. (2014). Multiple-documents literacy: Strategic processing, source awareness, and argumentation when reading multiple conflicting documents. *Learning and Individual Differences*, 30, 64–76. doi:10.1016/j.lindif.2013.01.007

Ariasi, N., & Mason, L. (2011). Uncovering the effect of text structure in learning from a science text: An eye-tracking study. *Instructional Science*, 39, 581–601. doi:10.1007/s11251-010-9142-5

Azevedo, R. (2009). Theoretical, conceptual, methodological, and instructional issues in research on metacognition and self-regulated learning: A discussion. *Metacognition and Learning*, 4, 87–95. doi:10.1007/s11409-009-9035-7

Baddeley, A. (1992). Working memory. *Science*, 255, 556–559.

Bailin, S., Case, R., Coombs, J. R., & Daniels, L. B. (1999). Common misconceptions of critical thinking. *Journal of Curriculum Studies*, 31(3), 269–283.

Beattle, V., Collins, B., & McInnes, B. (1997). Deep and surface learning: A simple or simplistic dichotomy? *Accounting Education: An International Journal*, 6, 1–12.

Beauregard, M., Lévesque, J., & Bourgouin, P. (2001). Neural correlates of conscious self-regulation of emotion. *Journal of Neuroscience*, 21, 1–6.

Biggs, J. B. (1987). *The Study Process Questionnaire manual*. Melbourne: Australian Council for Educational Research.

Binet, A., Simon, T., & Town, C. H. (1913). *A method of measuring the development of the intelligence of young children*. Chicago: Chicago Medical Book.

Black, P., & Wiliam, D. (1998). Inside the black box: Raising standards through classroom assessment. *Phi Delta Kappan*, 80(2), 139–148.

Block, R. A. (2009). Intent to remember briefly presented human faces and other pictorial stimuli enhances recognition memory. *Memory and Cognition*, 37, 667–678. doi:10.3758/MC.37.5.667

Boekaerts, M., Pintrich, P. R., & Zeidner, M. (2000). *Handbook of self-regulation*. San Diego, CA: Academic Press.

Bohan, J., & Sanford, A. (2008). Semantic anomalies at the borderline of consciousness: An eye-tracking investigation. *The Quarterly Journal of Experimental Psychology*, 61(2), 232–239. doi:10.1080/17470210701617219

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Bohn, R. E., & Short, J. E. (2009). *How much information? 2009 report on American consumers*. La Jolla, CA: UC San Diego Global Information Industry Center. Retrieved from http://group47.com/HMI_2009_ConsumerReport_Dec9_2009.pdf Accessed December 29, 2017.

Bostrom, A. (2008). Lead is like mercury: Risk comparisons, analogies and mental models. *Journal of Risk Research*, 11(1-2), 99-117. doi:10.1080/13669870701602956

Braasch, J. L. G., & Goldman, S. R. (2010). The role of prior knowledge in learning from analogies in science texts. *Discourse Processes*, 47, 447-479. doi: 10.1080/01638530903420960

Bråten, I., Ferguson, L. E., Anmarkrud, Ø., & Strømsø, H. I. (2013). Prediction of learning and comprehension when adolescents read multiple texts: The roles of word-level processing, strategic approach, and reading motivation. *Reading and Writing*, 26(3), 321-348.

Bråten, I., & Strømsø, H. I. (2003). A longitudinal think-aloud study of spontaneous strategic processing during the reading of multiple expository texts. *Reading and Writing*, 16, 195-218. doi:10.1023/A:1022895207490

Bråten, I., & Strømsø, H. I. (2006). Epistemological beliefs, interest, and gender as predictors of Internet-based learning activities. *Computers in Human Behavior*, 22, 1027-1042. doi:10.1016/j.chb.2004.03.026

Bråten, I., & Strømsø, H. I. (2011). Measuring strategic processing when students read multiple texts. *Metacognition & Learning*, 6, 111-130.

Bråten, I., Strømsø, H. I., & Britt, M. A. (2009). Examining the role of source evaluation in students' construction of meaning within and across multiple texts. *Reading Research Quarterly*, 44(1), 6-28.

Broughton, S. H., Sinatra, G. M., & Reynolds, R. E. (2010). The nature of the refutation text effect: An investigation of attention allocation. *The Journal of Educational Research*, 103(6), 407-423. doi:10.1080/00220670903383101

Butler, D. (1995). Promoting strategic learning by postsecondary students with learning disabilities. *Journal of Learning Disabilities*, 28(3), 170-190.

Cacioppo, J. T., & Petty, R. E. (1982). The need for cognition. *Journal of Personality and Social Psychology*, 42(1), 116-131.

Campione, J. C., & Brown, A. L. (1978). Toward a theory of intelligence: Contributions from research with retarded children. *Intelligence*, 2(3), 279-304.

Cano, F. (2007). Approaches to learning and study orchestrations in high school students. *European Journal of Psychology of Education*, 22, 131-151. doi:10.1007/BF03173518

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Case, J., & Marshall, D. (2004). Between deep and surface: Procedural approaches to learning in engineering education contexts. *Studies in Higher Education*, 29, 605–615. doi:10.1080/0307507042000261571

Cates, G. L., Burns, M. K., & Joseph, L. M. (2010). Introduction to the special issue: Instructional efficiency and the impact on learning and data-based decision making. *Psychology in the Schools*, 47, 111–113.

Cerdán, R., & Vidal-Abarca, E. (2008). The effects of tasks on integrating information from multiple documents. *Journal of Educational Psychology*, 100, 209–222. doi:10.1037/0022-0663.100.1.209

Chen, C., Zhou, X., Chen, C., Dong, Q., Zang, Y., Qiao, S., . . . Gong, Q. (2007). The neural basis of processing anomalous information. *NeuroReport: For Rapid Communication of Neuroscience Research*, 18(8), 747–751. doi:10.1097/WNR.0b013e3280ebb49b

Chi, M. T. H. (1985). Interactive roles of knowledge and strategies in the development of organized sorting and recall. In S. F. Chipman, J. W. Segal, & R. Glaser (Eds.), *Thinking and learning skills* (Vol. 2, pp. 457–484). Hillsdale, NJ: Erlbaum.

Chi, M. T. H., & Roscoe, R. D. (2002). The processes and challenges of conceptual change. In M. Limon & L. Mason (Eds.), *Reconsidering conceptual change: Issues in theory and practice* (pp. 3–27) Amsterdam: Kluwer.

Chin, C., & Brown, D. E. (2000). Learning in science: A comparison of deep and surface approaches. *Journal of Research in Science Teaching*, 37, 109–138.

Chinn, C. A., & Malhotra, B. A. (2002). Children's responses to anomalous scientific data: How is conceptual change impeded? *Journal of Educational Psychology*, 94(2), 327–343. doi:10.1037/0022-0663.94.2.327

Chinnappin, M., & Lawson, M. J. (1996). The effects of training in the use executive strategies in geometry problem solving. *Learning and Instruction*, 6, 1–17. doi:10.1016/S0959-4752(96)80001-6

Cho, S., Ryali, S., Geary, D. C., & Menon, V. (2011). How does a child solve $7 + 8$? Decoding brain activity patterns associated with counting and retrieval strategies. *Developmental Science*, 14, 989–1001. doi:10.1111/j.1467-7687.2011.01055.x

Coiro, J. (2003). Reading comprehension on the Internet: Expanding our understanding of reading comprehension to encompass new literacies. *The Reading Teacher*, 56(5), 458–464.

Coiro, J. (2011). Predicting reading comprehension on the Internet: Contributions of offline reading skills, online reading skills, and prior knowledge. *Journal of Literacy Research*, 43(4), 352–392.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Coiro, J., & Dobler, E. (2007). Exploring the online reading comprehension strategies used by sixth-grade skilled readers to search for and locate information on the Internet. *Reading Research Quarterly*, 42, 214–257. doi:10.1598/RRQ.42.2.2

Cole, M., & Wertsch, J. V. (1996). Beyond the individual-social antinomy in discussions of Piaget and Vygotsky. *Human Development*, 39(5), 250–256. doi:10.1159/000278475

Common Core State Standards Initiative (CCSS). (2010). *Common core state standards for English language arts and literacy in history/social studies, science, and technical subjects*. Washington, DC: Council of Chief State School Officers and National Governors Association. Retrieved from <http://www.corestandards.org/ELA-Literacy/> Accessed December 29, 2017.

Craik, F. I. M., & Lockhart, E. (1972). Levels of processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behavior*, 11, 671–684. doi:10.1016/S0022-5371(72)80001-X

Craik, F. I. M., & Tulving, E. (1975). Depth of processing and the retention of words in episodic memory. *Journal of Experimental Psychology: General*, 104, 268–294. doi:10.1037/0096-3445.104.3.268

Cromley, J. G., & Azevedo, R. (2006). Self-report of reading comprehension strategies: What are we measuring? *Metacognition and Learning*, 1, 229–247.

Darden, L. (1995). Exemplars, abstractions, and anomalies: Representations and theory change in Mendelian and molecular genetics. In J. G. Lennox & G. Wolters (Eds.), *Concepts, theories, and rationality in the biological sciences* (pp. 137–158). Pittsburgh, PA: University of Pittsburgh Press.

De La Paz, S. (2005). Effects of historical reasoning instruction and writing strategy master in culturally and academically diverse middle school classrooms. *Journal of Educational Psychology*, 97(2), 139–156.

Dermitzaki, I., Leondari, A., & Goudas, M. (2009). Relations between young children's strategic behaviours, domain-specific self-concept, and performance in a problem-solving situation. *Learning and Instruction*, 19, 144–157. doi:10.1016/j.learninstruc.2008.03.002

De Saussure, F. (2011). *Course in general linguistics* (W. Baskin, Trans.). New York: Columbia University Press. (Original work published 1916)

Dinsmore, D. L., & Alexander, P. A. (2012). A critical discussion of deep and surface processing: What it means, how it is measured, the role of context, and model specification. *Educational Psychology Review*, 24, 499–567.

Dinsmore, D. L., Alexander, P. A., & Loughlin, S. M. (2008). Focusing the conceptual lens on metacognition, self-regulation, and self-regulated learning. *Educational Psychology Review*, 20(4), 391–409.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Dresler, T., Obersteiner, A., Schecklmann, M., Carina A., Vogel, M., Ehli, A.-C., . . .

Fallgatter, A. J. (2009). Arithmetic tasks in different formats and their influence on behavior and brain oxygenation as assessed with near-infrared spectroscopy (NIRS): A study involving primary and secondary school children. *Journal of Neural Transmission*, 116, 1689–1700. doi:10.1007/s00702-009-0307-9

Duffy, G. G., & Roehler, L. R. (1989). Why strategy instruction is so difficult and what we need to do about it. In *Cognitive strategy research: From basic research to educational applications* (pp. 133–154). New York: Springer-Verlag.

Dumas, D., Alexander, P. A., & Grossnickle, E. M. (2013). Relational reasoning and its manifestations in the educational context: A systematic review of the literature. *Educational Psychology Review*, 29(3), 391–427.

Dunbar, K. (1995). How scientists really reason: Scientific reasoning in real-world laboratories. In R. J. Sternberg & J. E. Davidson (Eds.), *The nature of insight* (pp. 365–395). Cambridge, MA: MIT Press.

Dunlosky, J., & Hertzog, C. (2001). Measuring strategy production during associative learning: The relative utility of concurrent versus retrospective reports. *Memory and Cognition*, 29, 247–253. doi:10.3758/BF03194918

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, 14, 4–58. doi:10.1177/1529100612453266

Edwards Olshavsky, J. (1976). Reading as problem solving: An investigation of strategies. *Reading Research Quarterly*, 12, 654–674.

Entwhistle, N. J. (1991). Approaches to learning and perceptions of the learning environment. *Higher Education*, 22, 201–204.

Ericsson, K. A., Krampe, R. T., & Tesch-Römer, C. (1993). The role of deliberate practice in the acquisition of expert performance. *Psychological Review*, 100, 363–406.

Ericsson, K. A., & Simon, H. A. (1980). Verbal reports as data. *Psychological Review*, 87(3), 215–251.

Ericsson, K. A., & Simon, H. A. (1998). How to study thinking in everyday life: Contrasting think-aloud protocols with descriptions and explanations of thinking. *Mind, Culture, and Activity*, 5(3), 178–186.

Eshel, Y., & Kohavi, R. (2003). Perceived classroom control, self-regulated learning strategies, and academic achievement. *Educational Psychology*, 23, 249–260. doi:10.1080/0144341032000060093

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Ferguson, H. J., & Sanford, A. J. (2008). Anomalies in real and counterfactual worlds: An eye-movement investigation. *Journal of Memory and Language*, 58(3), 609–626. doi: 10.1016/j.jml.2007.06.007

Flavell, J. H. (1979). Metacognition and cognitive monitoring. *American Psychologist*, 34, 906–911.

Flavell, J. H. (1981). *Cognitive development*. Englewood Cliffs, NJ: Prentice-Hall.

Flavell, J. H., Friedrichs, A. G., & Hoyt, J. D. (1970). Developmental changes in memorization processes. *Cognitive Psychology*, 1, 324–340. doi: 10.1016/0010-0285(70)90019-8

Fox, E. (2009). The role of reader characteristics in processing and learning from informational text. *Review of Educational Research*, 79(1), 197–261.

Fox, E., & Riconscente, M. (2008). Metacognition and self-regulation in James, Piaget, and Vygotsky. *Educational Psychology Review*, 20, 373–389. doi:10.1007/s10648-008-9079-2

Fox, M. C., Ericsson, K. A., & Best, R. (2011). Do procedures for verbal reporting of thinking have to be reactive? A meta-analysis and recommendations for best reporting methods. *Psychological Bulletin*, 137, 316–344. doi:10.1037/a0021663

Freed, D. B., Marshall, R. C., & Phillips, D. S. (1998). Comparison of semantically and phonemically based training procedures in an overlearned naming task. *Perceptual and Motor Skills*, 87, 795–800.

Fuchs, L. S., Fuchs, D., Prentice, K., Burch, M., Hamlett, C. L., Owen, R., & Schroeter, K. (2003). Enhancing third-grade students' mathematical problem solving with self-regulated learning strategies. *Journal of Educational Psychology*, 95, 306–315. doi: 10.1037/0022-0663.95.2.306

Gardner, H. (1995). Perennial antinomies and perpetual redrawings: Is there progress in the study of mind? In R. Solso & D. Massaro (Eds.), *The science of the mind: 2001 and beyond* (pp. 65–78). New York: Oxford University Press.

Garner, R. (1982). Verbal-report data on reading strategies. *Journal of Literacy Research*, 14, 159–167. doi:10.1080/10862968209547443

Garner, R. (1987). *Metacognition and reading comprehension*. Norwood, NJ: Ablex.

Garner, R. (1990). When children and adults do not use learning strategies: Toward a theory of settings. *Review of Educational Research*, 60(4), 517–529.

Gentner, D., Loewenstein, J., & Thompson, L. (2003). Learning and transfer: A general role for analogical encoding. *Journal of Educational Psychology*, 95(2), 393–405. doi: 10.1037/0022-0663.95.2.393

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

- Glaser, R. (1984). Education and thinking: The role of knowledge. *American Psychologist*, 39, 93–104.
- Goswami, U. (1992). Analogical reasoning in children. In J. Campione, K. Metz, & A. S. Palincsar (Eds.), *Children's learning in laboratory and classroom contexts: Essays in honor of Ann Brown* (pp. 55–69). New York: Routledge.
- Grabner, R. H., Ansari, D., Koschutnig, K., Reishofer, G., Ebner, F., & Neuper, C. (2009). To retrieve or to calculate? Left angular gyrus mediates the retrieval of arithmetic facts during problem solving. *Neuropsychologia*, 47, 604–608. doi:10.1016/j.neuropsychologia.2008.10.013
- Grabner, R. H., & De Smedt, B. (2012). Oscillatory EEG correlates of arithmetic strategies: A training study. *Frontiers in Psychology*, 3, 1–11. doi:10.3389/fpsyg.2012.00428
- Graham, S., Harris, K. R., & Mason, L. H. (2005). Improving the writing performance, knowledge, and self-efficacy of struggling young writers: The effects of self-regulated strategy development. *Contemporary Educational Psychology*, 30, 207–241. doi:10.1016/j.cedpsych.2004.08.001
- Grodsky, A., & Giambra, L. M. (1990). The consistency across vigilance and reading tasks of individual differences in the occurrence of task-unrelated and task-related images and thoughts. *Imagination, Cognition and Personality*, 10, 39–52. doi:10.2190/6QG5-CXVV-4XUR-7P3K.
- Guinee, K., Eagleton, M. B., & Hall, T. E. (2003). Adolescents' Internet search strategies: Drawing upon familiar cognitive paradigms when accessing electronic information sources. *Journal of Educational Computing Research*, 29(3), 363–374.
- Guzenhaus, M. G. (2003). High stakes testing and the default philosophy of education. *Theory Into Practice*, 42, 51–58. doi:10.1207/s15430421tip4201_7
- Hagan, Å., Braasch, J. L. G., & Bråten, I. (2014). Relationships between spontaneous note-taking, self-reported strategies and comprehension when reading multiple texts in different task conditions. *Journal of Research in Reading*, 37(S1), S141–S157. doi:10.1111/j.1467-9817.2012.01536.x
- Hannus, M., & Hyönä, J. (1999). Utilization of illustrations during learning of science textbook passages among low- and high-ability children. *Contemporary Educational Psychology*, 24, 95–123. doi:10.1006/ceps.1998.0987
- Harris, K. R., Alexander, P. A., & Graham, S. (2008). Defining strategy research: Michael Pressley's contributions and directions for future research, *Educational Psychologist*, 43, 86–96.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Hatta, T., Kanari, A., Mase, M., Nagano, Y., Shirataki, T., & Hibino, S. (2009). Strategy effects on word searching in Japanese letter fluency tests: Evidence from the NIRS findings. *Reading and Writing*, 22, 1041–1052. doi:10.1007/s11145-008-9143-9

Heit, E., & Nicholson, S. P. (2010). The opposite of Republican: Polarization and political categorization. *Cognitive Science*, 34(8), 1503–1516. doi:10.1111/j.1551-6709.2010.01138.x

Hesse, M. B. (1959). On defining analogy. *Proceedings of the Aristotelian Society*, 60, 79–100.

Hidi, S., & Renninger, K. A. (2006). The four-phase model of interest development. *Educational Psychologist*, 41(2), 111–127.

Hofstadter, D. R. (2001). Epilogue: Analogy as the core of cognition. In D. Gentner, K. J. Holyoak, & B. N. Kokinov (Eds.), *The analogical mind: Perspectives from cognitive science* (pp. 499–538). Cambridge, MA: MIT Press.

Hofstadter, D. R., & Sander, E. (2013). *Surfaces and essences: Analogy as the fuel and fire of thinking*. New York: Basic Books.

Holliway, D. R. (2007). Spontaneous analogies in referential writing. *Communication & Cognition*, 40, 127–158.

Holyoak, K. J. (2012). Analogy and relational reasoning. In K. J. Holyoak & R. G. Morrison (Eds.), *The Oxford handbook of thinking and reasoning* (pp. 234–259). New York: Oxford University Press.

Hsiao, T.-Y., & Oxford, R. L. (2002). Comparing theories of language learning strategies: A confirmatory factor analysis. *Modern Languages Journal*, 86, 368–383. doi:10.1111/1540-4781.00155

Hulleman, C. S., & Harackiewicz, J. M. (2009). Promoting interest and performance in high school science classes. *Science*, 326, 1410–1412.

Hyönä, J., Lorch, J. F., Jr., & Kaakinen, J. K. (2002). Individual differences in reading to summarize expository text: Evidence from eye fixation patterns. *Journal of Educational Psychology*, 94, 44–55. doi:10.1037//0022-0663.94.1.44

James, W. (1890). *Principles of psychology* (Vols. 1 & 2). New York: Holt.

Jitendra, A. K., Star, J. R., Starosta, K., Leh, J. M., Sood, S., Caskie, G., . . . Mack, T. R. (2009). Improving seventh grade students' learning of ratio and proportion: The role of schema-based instruction. *Contemporary Educational Psychology*, 34, 250–264.

Just, M. A., & Carpenter, P. A. (1976). The role of eye-fixation research in cognitive psychology. *Behavior Research Methods & Instrumentation*, 8, 139–143. doi:10.3758/BF03201761

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

- Just, M. A., & Carpenter, P. A. (1980). A theory of reading: From eye fixations to comprehension. *Psychological Review*, 87, 329–354. doi:10.1037/0033-295X.87.4.329
- Kaakinen, J. K., & Hyönä, J. (2005). Perspective effects on expository text comprehension: Evidence from think-aloud protocols, eyetracking, and recall. *Discourse Processes*, 40, 239–257. doi:10.1207/s15326950dp4003_4
- Kaplan, A. S., & Murphy, G. L. (2000). Category learning with minimal prior knowledge. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 26(4), 829–846.
- Kapur, S., Craik, F. I. M., Tulving, E., Wilson, A. A., Houle, S., & Brown, G. M. (1994). Neuroanatomical correlates of encoding in episodic memory: Levels of processing effect. *Proceedings of the National Academy of Sciences of the United States of America*, 91, 2008–2011.
- Keys, C. W., Hand, B., Prain, V., & Collins, S. (1999). Using the science writing heuristic as a tool for learning from laboratory investigations in secondary science. *Journal of Research in Science Teaching*, 36(10), 1065–1084.
- King, P. M., & Kitchener, K. S. (2002). The reflective judgment model: Twenty years of research on epistemic cognition. In B. K. Hofer & P. R. Pintrich (Eds.), *Personal epistemology: The psychology of beliefs about knowledge and knowing* (pp. 37–61). Mahwah, NJ: Erlbaum.
- Kjeldergaard, P. M., & Higa, M. (1962). Degree of polarization and the recognition value of words selected from the semantic atlas. *Psychological Reports*, 11(3), 629–630. doi:10.2466/pr0.1962.11.3.629
- Klinger, J. K., Vaughn, S., & Schumm, J. S. (1998). Collaborative strategic reading during social studies in heterogeneous fourth-grade classrooms. *The Elementary School Journal*, 99(1), 3–22.
- Krawczyk, D. C. (2012). The cognition and neuroscience of relational reasoning. *Brain Research*, 1428, 13–23. doi:10.1016/j.brainres.2010.11.080
- Kuhn, D., & Udell, W. (2007). Coordinating own and other perspectives in argument. *Thinking & Reasoning*, 13(2), 90–104. doi:10.1080/13546780600625447
- Kuhn, T. S. (1996). *The structure of scientific revolutions*. Chicago: University of Chicago Press.
- Kulturel-Konak, S., Konak, A., Esparragoza, I. E., & Kremer, G. E. O. (2013, January). Assessing professional skills in STEM disciplines. In *Proceedings of the 2013 IEEE Integrated STEM Education Conference* (pp. 1–4), Princeton, NJ.
- Kurby, C. A., Britt, M. A., & Magliano, J. P. (2005). The role of top-down and bottom-up processes in between-text integration. *Reading Psychology*, 26, 335–362.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Langan, D., & Athanasou, J. (2005). Testing a model of domain learning in music therapy. *Journal of Music Therapy*, 42(4), 296–312.

Leu, D., Zawilinski, L., Castek, J., Banerjee, M., Housand, B., & Liu, Y. (2008). What is new about the new literacies of online reading comprehension? In L. S. Rush, A. J. Eakle, & A. Berger (Eds.), *Secondary school literacy: What research reveals for classroom practices* (pp. 37–68). Urbana, IL: National Council of Teachers of English.

Linnenbrink-Garcia, L., & Pekrun, R. (2011). Students' emotions and academic engagement: Introduction to the special issue. *Contemporary Educational Psychology*, 36(1), 1–3.

List, A., Grossnickle, E. M., & Alexander, P. A. (2012, September). *Students' source selections, justifications, and evaluations when responding to different question types*. Presented at the International Conference on Conceptual Change, Trier, Germany.

Litman, J. A. (2010). Relationships between measures of I- and D-type curiosity, ambiguity tolerance, and need for closure: An initial test of the wanting-liking model of information-seeking. *Personality and Individual Differences*, 48, 397–402. doi:10.1016/j.paid.2009.11.005

Marková, I. (1987). On the interaction of opposites in psychological processes. *Journal for the Theory of Social Behaviour*, 17(3), 279–299. doi:10.1111/j.1468-5914.1987.tb00100.x

Marton, F., & Säljö, R. (1976a). On qualitative differences in learning I: Outcome and process. *British Journal of Educational Psychology*, 46, 4–11.

Marton, F., & Säljö, R. (1976b). On qualitative differences in learning II: Outcome as a function of the learner's conception of the task. *British Journal of Educational Psychology*, 46, 115–127.

Mason, L., & Ariasi, N. (2011). Uncovering the effect of text structure in learning from a science text: An eye-tracking study. *Instructional Science*, 39, 581–601. doi:10.1007/s11251-010-9142-5

Mason, L. H. (2004). Explicit self-regulated strategy development versus reciprocal questioning: Effects on expository reading comprehension among struggling readers. *Journal of Educational Psychology*, 96, 283–296. doi:10.1037/0022-0663.96.2.283

Mason, L. H. (2013). Teaching students who struggle with learning to think before, while, and after reading: Effects of self-regulated strategy development instruction. *Reading and Writing Quarterly*, 29, 124–144. doi:10.1080/10573569.2013.758561

Mason, L. H., Snyder, K. H., Sukhram, D. P., & Kedem, Y. (2006). TWA + PLANS strategies for expository reading and writing: Effects for nine fourth-grade students. *Exceptional Children*, 73(1), 69–89.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Mayer, R. E. (2011). Towards a science of motivated learning in technology-supported environments. *Educational Technology Research and Development*, 59(2), 301–308.

McCrudden, M. T., Magliano, J. P., & Schraw, G. (2010). Exploring how relevance instructions affect personal reading intentions, reading goals and text processing: A mixed methods study. *Contemporary Educational Psychology*, 25, 229–241. doi:10.1016/j.cedpsych.2009.12.001

McNeil, L. M. (2000). *Contradictions of school reform: Educational costs of standardized testing*. New York: Routledge.

Meneghetti, C., & De Beni, R. (2010). Influence of motivational beliefs and strategies on recall task performance in elementary, middle and high school students. *European Journal of Psychology of Education*, 25(3), 325–343. doi:10.1007/s10212-010-0019-4

Metzger, M. J., Flanagin, A. J., & Zwarun, L. (2003). College student web use, perceptions of information credibility, and verification behavior. *Computers & Education*, 41(3), 271–290. doi:10.1016/S0360-1315(03)00049-6

Moeller, K., Klein, E., & Nuerk, H.-C. (2011). (No) Small adults: Children's processing of carry addition problems. *Developmental Neuropsychology*, 36, 702–720. doi:10.1080/87565641.2010.549880

Moos, D. C., & Azevedo, R. (2008). Self-regulated learning with hypermedia: The role of prior domain knowledge. *Contemporary Educational Psychology*, 33, 270–298. doi:10.1016/j.cedpsych.2007.03.001

Mosenthal, P. B. (1988). Anopheles and antinomies in reading research. *Reading Teacher*, 42(3), 234–235.

Moss, J., Schunn, C. D., Schneider, W., McNamara, D. S., & VanLehn, K. (2011). The neural correlates of strategic reading comprehension: Cognitive control and discourse comprehension. *NeuroImage*, 58, 675–686. doi:10.1016/j.neuroimage.2011.06.034

Muis, K. R. (2007). The role of epistemic beliefs in self-regulated learning. *Educational Psychologist*, 42, 173–190.

Muis, K. R., Bendixen, L. D., & Haerle, F. C. (2006). Domain-generality and domain-specificity in personal epistemology research: Philosophical and empirical reflections in the development of a theoretical framework. *Educational Psychology Review*, 18, 3–54. doi:10.1007/s10648-006-9003-6

Muis, K. R., & Franco, G. M. (2009). Epistemic beliefs: Setting the standards for self-regulated learning. *Contemporary Educational Psychology*, 34, 306–318. doi:10.1016/j.cedpsych.2009.06.005

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Murayama, K., & Elliot, A. J. (2009). The joint influence of personal achievement goals and classroom goal structures on achievement-relevant outcomes. *Journal of Educational Psychology, 101*, 432–447. doi:10.1037/a0014221

Murphy, P. K., & Alexander, P. A. (2000). A motivated look at motivational terminology [Special issue]. *Contemporary Educational Psychology, 25*, 3–53.

Murphy, P. K., & Alexander, P. A. (2002). What counts: The predictive power of subject-matter knowledge, strategic processing, and interest in domain-specific performance. *Journal of Experimental Education, 70*, 197–214.

Murphy, P. K., Alexander, P. A., & Muis, K. R. (2012). *Knowledge and knowing: The journey from philosophy and psychology to human learning*. In K. R. Harris, S. Graham, & T. Urdan (Eds.), *Educational psychology handbook: Vol. 1. Theories, constructs, and critical issues* (pp. 189–226). Washington, DC: American Psychological Association.

Murphy, P. K., & Mason, L. (2006). Changing knowledge and beliefs. In P. A. Alexander & P. H. Winne (Eds.), *Handbook of educational psychology* (pp. 305–324). New York: Taylor & Francis.

Nett, U. T., Goetz, T., Hall, N. C., & Frenzel, A. C. (2012). Metacognitive strategies and test performance: An experience sampling analysis of students' learning behavior. *Education Research International, 2012*, 1–16.

Newman, S. D., Willoughby, G., & Price, B. (2011). The effect of problem structure on problem-solving: An fMRI study of word versus number problems. *Brain Research, 1410*, 77–88. doi:10.1016/j.brainres.2011.06.053

Nolen, S. B. (1988). Reasons for studying: Motivational orientations and study strategies. *Cognition and Instruction, 5*, 269–287.

Nolen, S. B., & Haladyna, T. M. (1990). Personal and environmental influences on students' beliefs about effective study strategies. *Contemporary Educational Psychology, 15*, 116–130. doi:10.1016/0361-476X(90)90011-O

Opfer, J. E., & Gelman, S. A. (2011). Development of the animate-inanimate distinction. In U. Goswami (Ed.), *The Wiley-Blackwell handbook of childhood cognitive development* (2nd ed., pp. 213–238). Malden, MA: Wiley-Blackwell.

Palincsar, A. S., & Brown, A. L. (1984). Reciprocal teaching of comprehension-fostering and comprehension-monitoring activities. *Cognition and Instruction, 1*(2), 117–175.

Pape, S. G., & Wang, C. (2003). Middle school children's strategic behavior: Classification and relation to academic achievement and mathematical problem solving. *Instructional Science, 31*, 419–449.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

- Patrick, H., Kaplan, A., & Ryan, A. M. (2011). Positive classroom motivational environments: Convergence between mastery goal structure and classroom social climate. *Journal of Educational Psychology, 103*, 367–382. doi:10.1037/a0023311
- Payne, S. J., & Reader, W. R. (2006). Constructing structure maps of multiple on-line texts. *International Journal of Human-Computer Studies, 64*, 461–474. doi:10.1016/j.ijhcs.2005.09.003
- Pekrun, R., Goetz, T., Titz, W., & Perry, R. P. (2002). Academic emotions in students' self-regulated learning and achievement: A program of qualitative and quantitative research. *Educational Psychologist, 37*(2), 91–105.
- Petty, R. E., Briñol, P., Loersch, C., & McCaslin, M. J. (2009). The need for cognition. In M. R. Leary & R. H. Hoyle (Eds.), *Handbook of individual differences in social behavior* (pp. 318–329). New York: Guilford Press.
- Phan, H. P. (2009). Exploring students' reflective thinking practice, deep processing strategies, effort, and achievement goal orientations. *Educational Psychology, 29*, 297–313. doi:10.1080/01443410902877988
- Pintrich, P. R. (1991). Editor's comment. *Educational Psychologist, 26*(3-4), 199–205.
- Pintrich, P. R. (2000). The role of goal orientation in self-regulated learning. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation* (pp. 451–502). San Diego, CA: Academic Press.
- Pintrich, P. R., & De Groot, E. V. (1990). Motivational and self-regulated learning components of classroom academic performance. *Journal of Educational Psychology, 82*, 33–40.
- Pintrich, P. R., Smith, D. A., Garcia, T., & McKeachie, W. J. (1993). Reliability and predictive validity of the Motivated Strategies for Learning Questionnaire (MSLQ). *Educational and Psychological Measurement, 53*(3), 801–813. doi:10.1177/0013164493053003024
- Pokay, P., & Blumenfeld, P. C. (1990). Predicting achievement early and late in the semester: The role of motivation and use of learning strategies. *Journal of Educational Psychology, 82*, 41–50. doi:10.1037/0022-0663.82.1.41
- Pressley, M., & Afflerbach, P. (1995). *Verbal protocols of reading: The nature of constructively responsive reading*. New York: Routledge.
- Pressley, M., Goodchild, F., Fleet, J., Zajchowski, R., & Evans, E. D. (1989). The challenges of classroom strategy instruction. *The Elementary School Journal, 89*(3), 301–342.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Pressley, M., Johnson, C. J., Symons, S., McGoldrick, J. A., & Kurita, J. A. (1989). Strategies that improve children's memory and comprehension of text. *The Elementary School Journal*, 90, 3-32.

Purpura, J. E. (1997). An analysis of the relationships between test takers' cognitive and metacognitive strategy use and second language test performance. *Language Learning*, 47, 289-325. doi:10.1111/0023-8333.91997009

Ramsden, P. (1997). The context of learning in academic departments. In F. Marton, D. Hounsell, & N. J. Entwistle (Eds.), *The experience of learning: Implications for teaching and studying in higher education* (pp. 198-216). Edinburgh: Scottish Academic Press.

Rasmussen, T. (2007). *Hypertext reading as practical action: Notes on technology, objectivation and knowledge*. Retrieved from http://www.wiumlie.no/1993/teletronikk-4-93/Rasmussen_T.html. Accessed December 29, 2017.

Rayner, K. (1978). Eye movements in reading and information processing. *Psychological Bulletin*, 85, 618-660.

Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin*, 124, 372-422.

Rayner, K., Chance, K. H., Slatterby, T. J., & Ashby, J. (2006). Eye movements as reflections of comprehension processes in reading. *Scientific Studies of Reading*, 10, 241-255. doi:10.1207/s1532799xssr1003_3

Resnick, L. B. (1987). *Education and learning to think*. Washington, DC: National Academies Press.

Richland, L. E., & McDonough, I. M. (2010). Learning by analogy: Discriminating between potential analogs. *Contemporary Educational Psychology*, 35(1), 28-43. doi:10.1016/j.cedpsych.2009.09.001

Richland, L. E., Zur, O., & Holyoak, K. J. (2007). Cognitive supports for analogies in the mathematics classroom. *Science*, 316(5828), 1128-1129. doi:10.1126/science.1142103

Richter, T., & Schmid, S. (2010). Epistemological beliefs and epistemic strategies in self-regulated learning. *Metacognition and Learning*, 5, 47-65. doi:10.1007/s11409-009-9038-4

Rideout, V. J., Foehr, U. G., & Roberts, D. F. (2010). *Generation M2: Media in the lives of 8- to 18-year-olds*. Menlo Park, CA: Kaiser Family Foundation.

Rodriguez, C. M. (2009). The impact of academic self-concept, expectations and the choice of learning strategy on academic achievement: The case of business students. *Higher Education Research and Development*, 28, 523-539. doi:10.1080/07294360903146841

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Rosenburg-Lee, M., Barth, M., & Menon, V. (2011). What difference does a year of schooling make? Maturation of brain response and connectivity between 2nd and 3rd grades during arithmetic problem solving. *NeuroImage*, 57, 796–808. doi:10.1016/j.neuroimage.2011.05.013

Russell, B., & Lackey, D. (1973). *Essays in analysis*. New York: Allen & Unwin.

Samuelstuen, M. S., & Bråten, I. (2007). Examining the validity of self-reports on scales measuring students' strategic processing. *British Journal of Educational Psychology*, 77, 351–378. doi:10.1348/1000709906X106147

Schellings, G. (2011). Applying learning strategy questionnaires: Problems and possibilities. *Metacognition and Learning*, 6, 91–109. doi:10.1007/s11409-011-9069-5

Schiff, R., & Ravid, D. (2007). Morphological analogies in Hebrew-speaking university students with dyslexia compared with typically developing gradeschoolers. *Journal of Psycholinguistic Research*, 36(3), 237–253. doi:10.1007/s10936-006-9043-6

Schmeck, R. R., Ribich, F., & Ramanaiah, N. (1977). Development of a self-report inventory for assessing individual differences in learning processes. *Applied Psychological Measurement*, 1, 413–431. doi:10.1177/014662167700100310

Schommer, M. (1993). Epistemological development and academic performance among secondary students. *Journal of Educational Psychology*, 85, 406–411.

Schrader, P. G., Lawless, K., & Mayall, H. (2008). The model of domain learning as a framework for understanding Internet navigation. *Journal of Educational Multimedia and Hypermedia*, 17(2), 235–258.

Schraw, G. (1998). Promoting general metacognitive awareness. *Instructional Science*, 26, 113–125.

Schraw, G., & Robinson, D. (Eds.). (2011). *Assessment of higher order thinking skills*. Charlotte, NC: Information Age.

Schunk, D. H. (2008). Metacognition, self-regulation, and self-regulated learning: Research recommendations. *Educational Psychology Review*, 20(4), 463–467.

Schwartz, M. C., Sadler, P. M., Sonnert, G., & Tai, R. H. (2009). Depth versus breadth: How content coverage in high school science courses relates to later success in college science coursework. *Science Education*, 93, 798–826.

Shanahan, T., & Shanahan, C. (2008). Teaching disciplinary literacy to adolescents: Rethinking content-area literacy. *Harvard Educational Review*, 78, 40–59.

Shaumyan, S. (2006). Antinomies of language and language operations of the mind. In H. R. Arabnia, E. B. Kozerenk, & S. Shaumyan. (Eds.), *Proceedings of the 2006 International*

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Conference on Machine Learning: Models, technologies and applications (pp. 3–9). Las Vegas, NV: CSREA Press.

Shen, B., & Chen, A. (2006). Examining the interrelations among knowledge, interests, and learning strategies. *Journal of Teaching in Physical Education*, 25, 182–199.

Silvia, P. J., & Sanders, C. E. (2010). Why are smart people curious? Fluid intelligence, openness to experience, and interest. *Learning and Individual Differences*, 20, 242–245. doi:10.1016/j.lindif.2010.01.006

Simon, W. G., & Chase, H. A. (1973). Perception in chess. *Cognitive Psychology*, 4, 55–61.

Sinatra, G. M., & Pintrich, P. R. (Eds.). (2002). *Intentional conceptual change*. New York: Routledge.

Slater, W. H., & Horstman, F. R. (2002). Teaching reading and writing to struggling middle school and high school students: The case for reciprocal teaching. *Preventing School Failure*, 46(4), 163–166.

Sorensen, R. A. (2003). *A brief history of the paradox: Philosophy and the labyrinths of the mind*. New York: Oxford University Press.

Springer, M. G., & Gardner, C. D. (2010). Teacher pay for performance: Context, status, and direction. *Phi Delta Kappan*, 91, 8–15.

Spörer, N., Brunstein, J., & Kieschke, U. (2009). Improving students' reading comprehension skills: Effects of strategy instruction and reciprocal teaching. *Learning and Instruction*, 19, 272–286. doi:10.1016/j.learninstruc.2008.05.003

Stanovich, K. E. (2009). *What intelligence tests miss: The psychology of rational thought*. New Haven, CT: Yale University Press.

Stanovich, K. E., West, R. F., & Toplak, M. E. (2012). Intelligence and rationality. In R. Sternberg & S. B. Kaufman (Eds.), *Cambridge handbook of intelligence* (3rd ed., pp. 784–826). Cambridge, England: Cambridge University Press.

Stein, B. L., & Kirby, J. R. (1992). The effects of text absent and text present conditions on summarization and recall of text. *Journal of Reading Behavior*, 24, 217–232. doi:10.1080/10862969209547773

Sternberg, R. J. (1977). *Intelligence, information processing, and analogical reasoning: The componential analysis of human abilities*. Oxford, England: Erlbaum Stevenson.

Strømsø, H. I., & Bråten, I. (2010). The role of personal epistemology in the self-regulation of Internet-based learning. *Metacognition and Learning*, 4, 91–111.

Swanson, H. L. (1989). Strategy instruction: Overview of principles and procedures for effective use. *Learning Disability Quarterly*, 12(1), 3–14.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

- Sweller, J. (1990). On the limited evidence for the effectiveness of teaching general problem-solving strategies. *Journal for Research in Mathematics Education*, 21, 411–414.
- Trey, L., & Khan, S. (2008). How science students can learn about unobservable phenomena using computer-based analogies. *Computers & Education*, 51(2), 519–529. doi:10.1016/j.compedu.2007.05.019
- Trickett, S. B., Trafton, J. G., & Schunn, C. D. (2009). How do scientists respond to anomalies? Different strategies used in basic and applied science. *Topics in Cognitive Science*, 1, 711–729. doi:10.1111/j.1756-8765.2009.01036.x
- Vakil, E., Lifshitz, H., Tzuriel, D., Weiss, I., & Arzuwan, Y. (2011). Analogies solving by individuals with and without intellectual disability: Different cognitive patterns as indicated by eye movements. *Research in Developmental Disabilities*, 32, 846–856. doi:10.1016/j.ridd.2010.08.006
- van Dijk, T. A., & Kintsch, W. (1983). *Strategies of discourse comprehension*. New York: Academic Press.
- van Dinther, M., Dochy, F., & Segers, M. (2011). Factors affecting students' self-efficacy in higher education. *Educational Research Review*, 6, 95–108. doi:10.1016/j.edurev.2010.10.003
- Veenman, M. V. J., Van-Hout Wolters, B. H. A. M., & Afflerbach, P. (2006). Metacognition and learning: Conceptual and methodological considerations. *Metacognition and Learning*, 1, 3–14. doi:10.1007/s11409-006-6893-0
- Verheij, J., Stoutjesdijk, E., & Beishuizen, J. (1996). Search and study strategies in hypertext. *Computers in Human Behavior*, 12, 1–15. doi:10.1016/0747-5632(95)00015-1
- Vigneau, F., Caissie, A. F., & Bors, D. A. (2006). Eye-movement analysis demonstrates strategic influences on intelligence. *Intelligence*, 34, 261–272. doi:10.1016/j.intell.2005.11.003
- Vrugt, A., & Oort, F. J. (2008). Metacognition, achievement goals, study strategies and academic achievement: Pathways to achievement. *Metacognition and Learning*, 3, 123–146. doi:10.1007/s11409-008-9022-4
- Walraven, A., Brand-Gruwel, S., & Boshuizen, H. (2009). How students evaluate information and sources when searching the World Wide Web for information. *Computers & Education*, 52(1), 234–246.
- Weinstein, C. E., Goetz, E. T., & Alexander, P. A. (Eds.). (1988). *Learning and study strategies: Issues in assessment, instruction, and evaluation*. San Diego, CA: Academic Press.

A Retrospective and Prospective Examination of Cognitive Strategies and Academic Development: Where Have We Come in Twenty-Five Years?

Weinstein, C. E., & Mayer, R. E. (1986). The teaching of learning strategies. In M. C. Wittrock (Ed.), *Handbook of research on teaching* (3rd ed., pp. 315–327). New York: Macmillan.

Weir, M. W. (1964). Developmental changes in problem-solving strategies. *Psychological Review*, 71, 473–490.

Wigfield, A., & Eccles, J. S. (2000). Expectancy-value theory of achievement motivation. *Contemporary Educational Psychology*, 25(1), 68–81.

Wineburg, S. S. (1991). Historical problem solving: A study of the cognitive processes used in the evaluation of documentary and pictorial evidence. *Journal of Educational Psychology*, 83, 73–87. doi:10.1037/0022-0663.83.1.73

Winne, P., & Perry, N. (2000). Measuring self-regulated learning. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation* (pp. 531–566). San Diego, CA: Academic Press.

Woessmann, L. (2011). Cross-country evidence on teacher performance pay. *Economics of Education Review*, 30, 404–418. doi:10.1016/j.econedurev.2010.12.008

Wolters, C. A., & Pintrich, P. R. (1998). Contextual differences in student motivation and self-regulated learning in mathematics, English, and social studies classrooms. *Instructional Science*, 26, 27–47. doi:10.1023/A:1003035929216

Yang, K.-L. (2012). Structures of cognitive and metacognitive reading strategy use for reading comprehension of geometry proof. *Educational Studies in Mathematics*, 80, 307–326. doi:10.1007/s10649-011-9350-1

Youssef, A., Ayers, P., & Sweller, J. (2012). Using general problem-solving strategies to generate ideas in order to solve geography problems. *Applied Cognitive Psychology*, 26, 872–877. doi:10.1002/acp.2888

Zimmerman, B. J. (1995). Self-regulation involves more than metacognition: A social cognitive perspective. *Educational Psychologist*, 30(4), 217–221.

Zimmerman, B. J. (2002). Self-regulated learning and academic achievement: An overview. *Educational Psychologist*, 25, 3–17. doi:10.1207/s15326985ep2501_2

Zimmerman, B. J., & Schunk, D. H. (Eds.). (2011). *Handbook of self-regulation of learning and performance*. New York: Taylor & Francis.

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